INTRODUCTION TO IMAGE PROCESSING

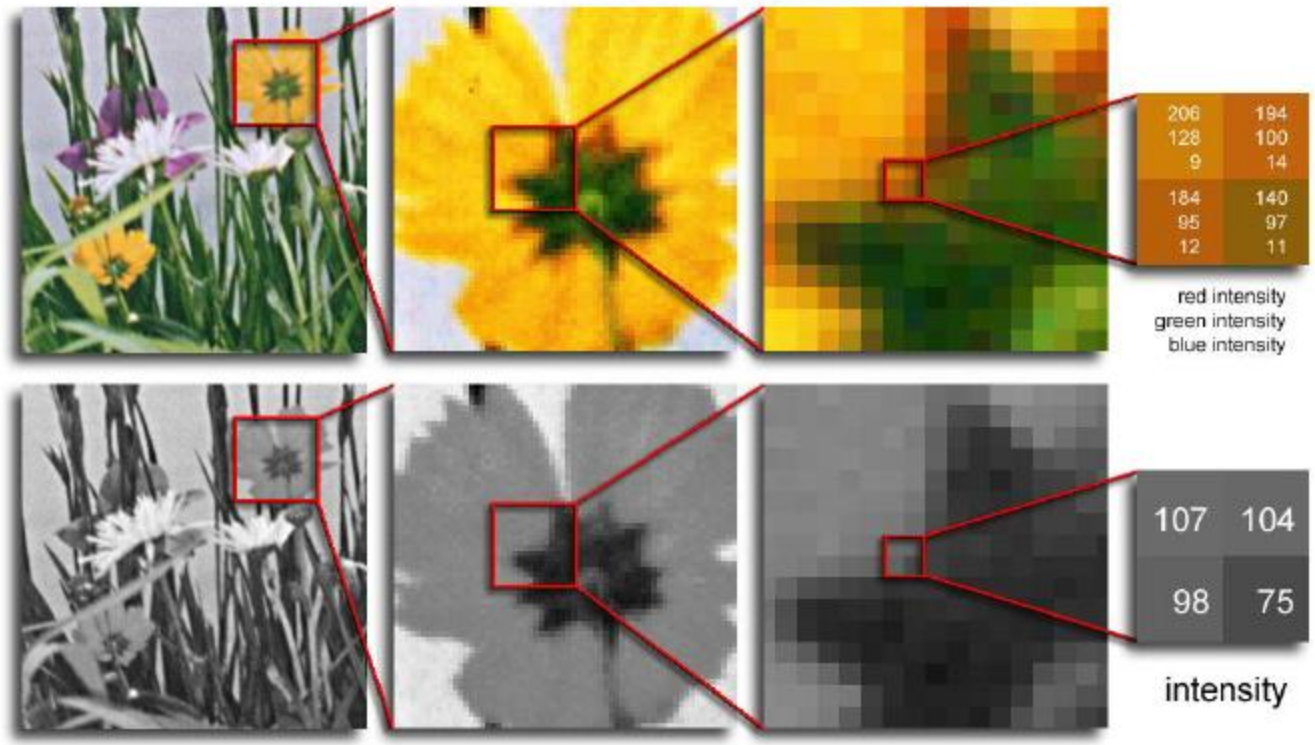
LECTURE 2
Image Basics
Image Enhancement in Spatial Domain

Digital Image

Color images have 3 values per pixel; monochrome images have 1 value per pixel.

a grid of squares, each of which contains a single color

each square is called a pixel (for *picture element*)



Digital 2D image (Grayscale)

Origin at (x, y) = (0, 0) in the textbook and for C, C++, Python, etc.

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

Origin at (x, y) = (1, 1) for MATLAB

$$f(x, y) = \begin{bmatrix} f(1,1) & f(1,2) & \ldots & f(1,N) \\ f(2,1) & f(2,2) & \ldots & f(2,N) \\ \ldots & \ldots & \ldots & \ldots \\ f(M,1) & f(M,2) & \ldots & f(M,N) \end{bmatrix}$$

f is a function that assigns a gray-level value (an integer between 0 and $L-1$) to each distinct pair of coordinates (x, y).

Spatial and Gray-Level Resolution

- Spatial resolution
  — A measure of the smallest discernible detail in an image
  — stated with *line pairs per unit distance, dots (pixels) per unit distance, dots per inch (dpi)*

- Gray-level (Intensity) resolution
  — The smallest discernible change in intensity level
  — stated with *8 bits, 16 bits, etc.*
    8 bits → $L = 256$ gray levels (most common)
    16 bits → $L = 65{,}536$ gray levels (medical images, etc.)

We refer to an L-level digital image of size $M \times N$ as having a spatial resolution of $M \times N$ pixels and a gray-level resolution of L levels.

Spatial Resolution



A 1024×1024, 8-bit image subsampled down to size 32×32 pixels. The number of allowable gray levels was kept at 256.

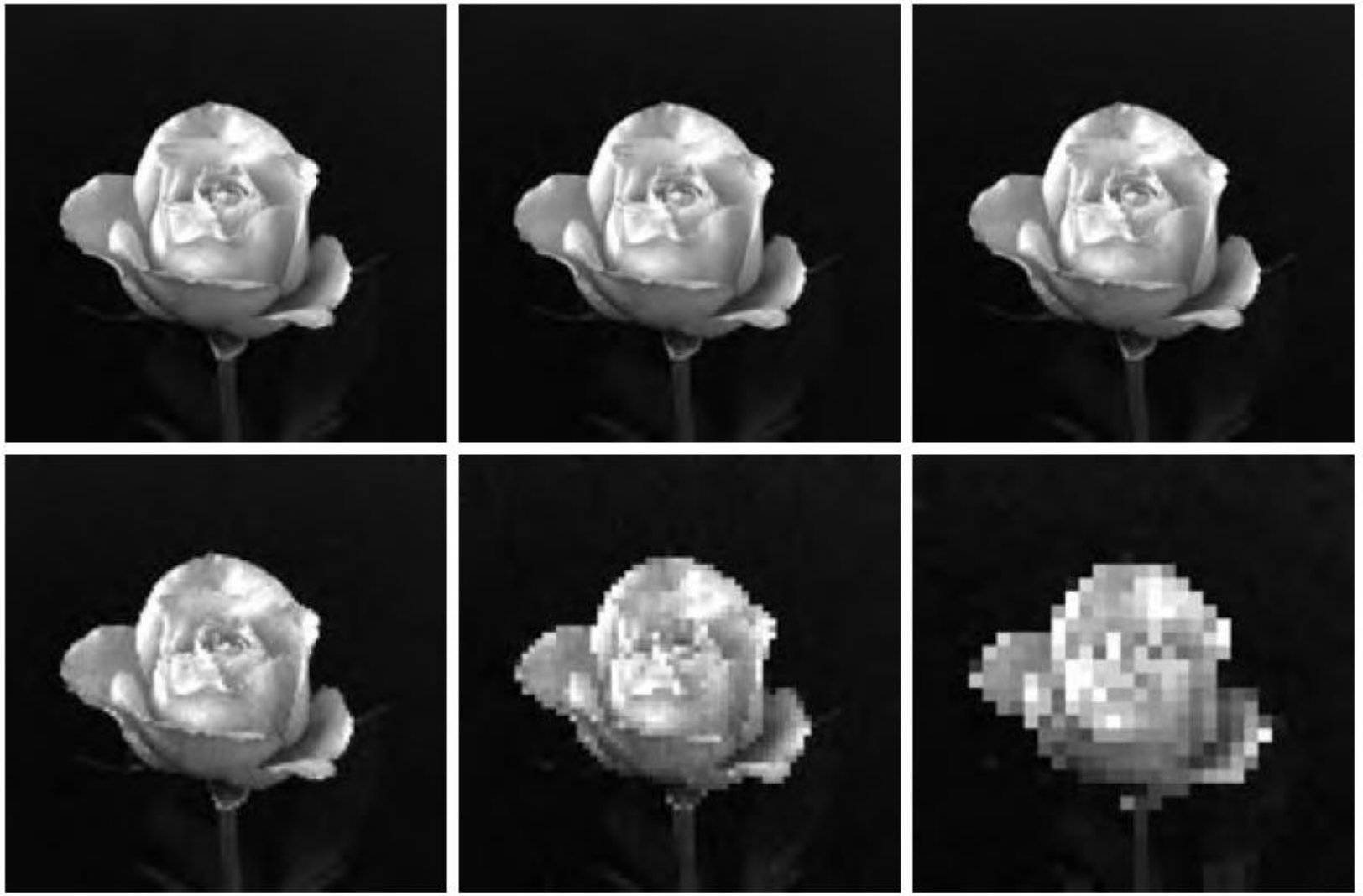
Spatial Resolution



The subsampling was accomplished by deleting the appropriate number of rows and columns from the original image.

For example, the 512×512 image was obtained by deleting every other row and column from the 1024×1024 image.

Spatial Resolution



As we reduce the spatial resolution from 1024×1024 down to 32×32:

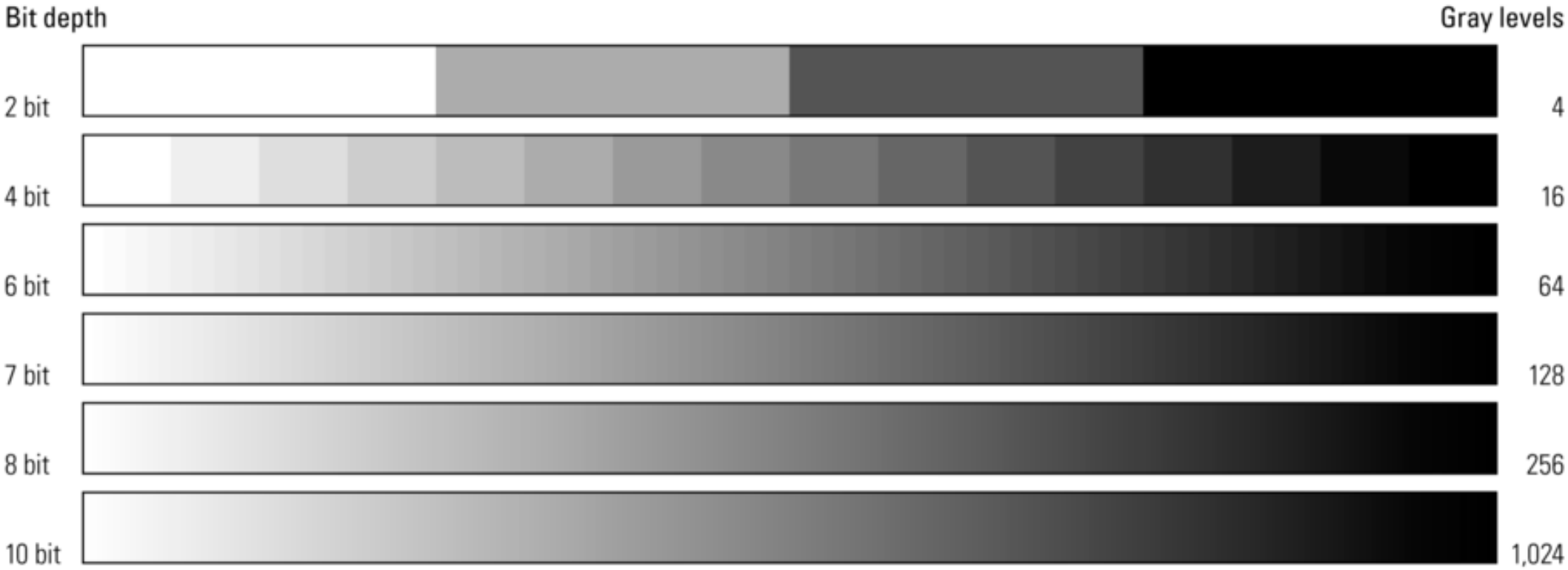
- The level of detail is reduced.
- The checker-board pattern becomes visible.

a b c
d e f

(a) 1024*1024, 8-bit image. (b) Zoomed 512*512 image. (c) through (f) 256*256, 128*128, 64*64, and 32*32 images (zoomed)

Gray-level (Intensity) Resolution

Gray-level resolution refers to the smallest discernible change in gray level (or intensity). It is the number of shades of gray utilized in the image for storage or display.



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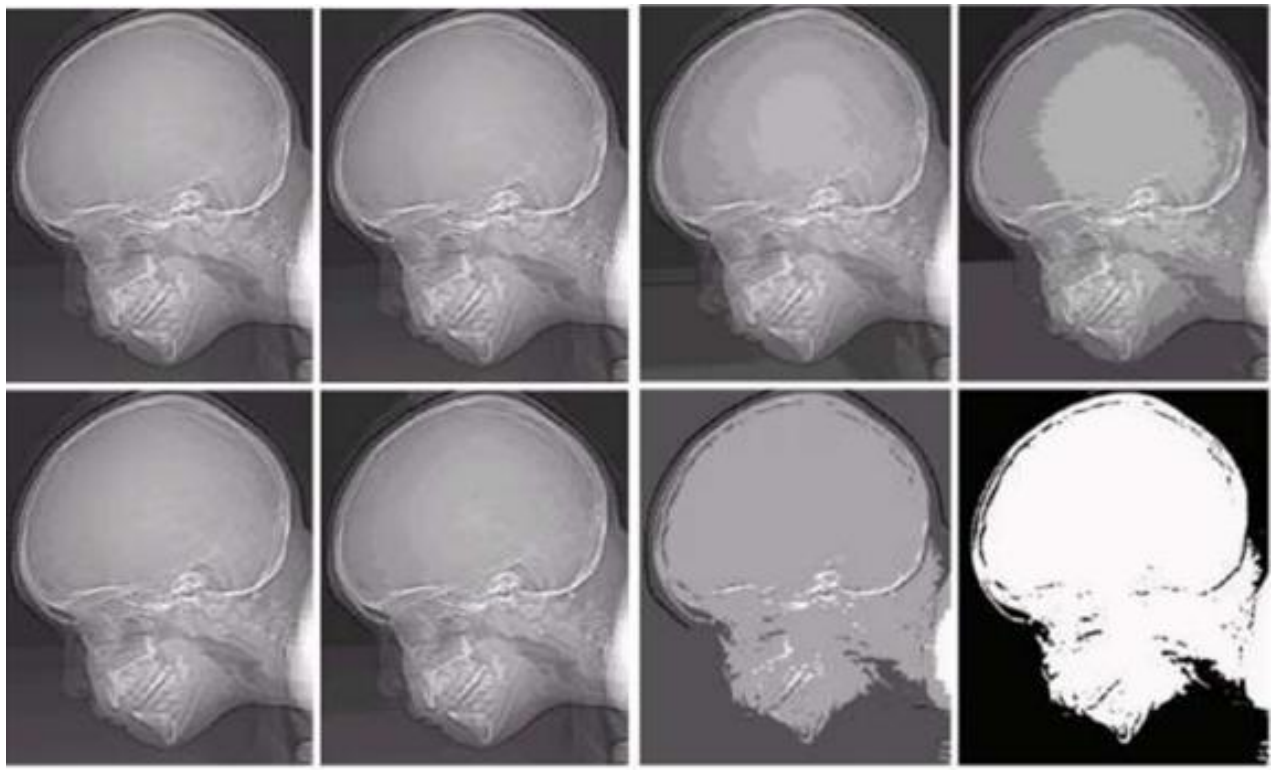


Image displayed in 256, 128, 64, 32, 16, 8, 4, 2 gray-levels

Gray-level (Intensity) Resolution

- With low gray-level resolution we observe ridgeline structurese in areas of smooth gray levels.
- This effect, caused by the use of an insufficient number of gray levels in smooth areas of a digital image, is called *false contouring*, so called because the ridges resemble topographic contours in a map.
- False contouring generally is quite visible in images displayed using 16 or less uniformly spaced gray levels.



256 gray-levels



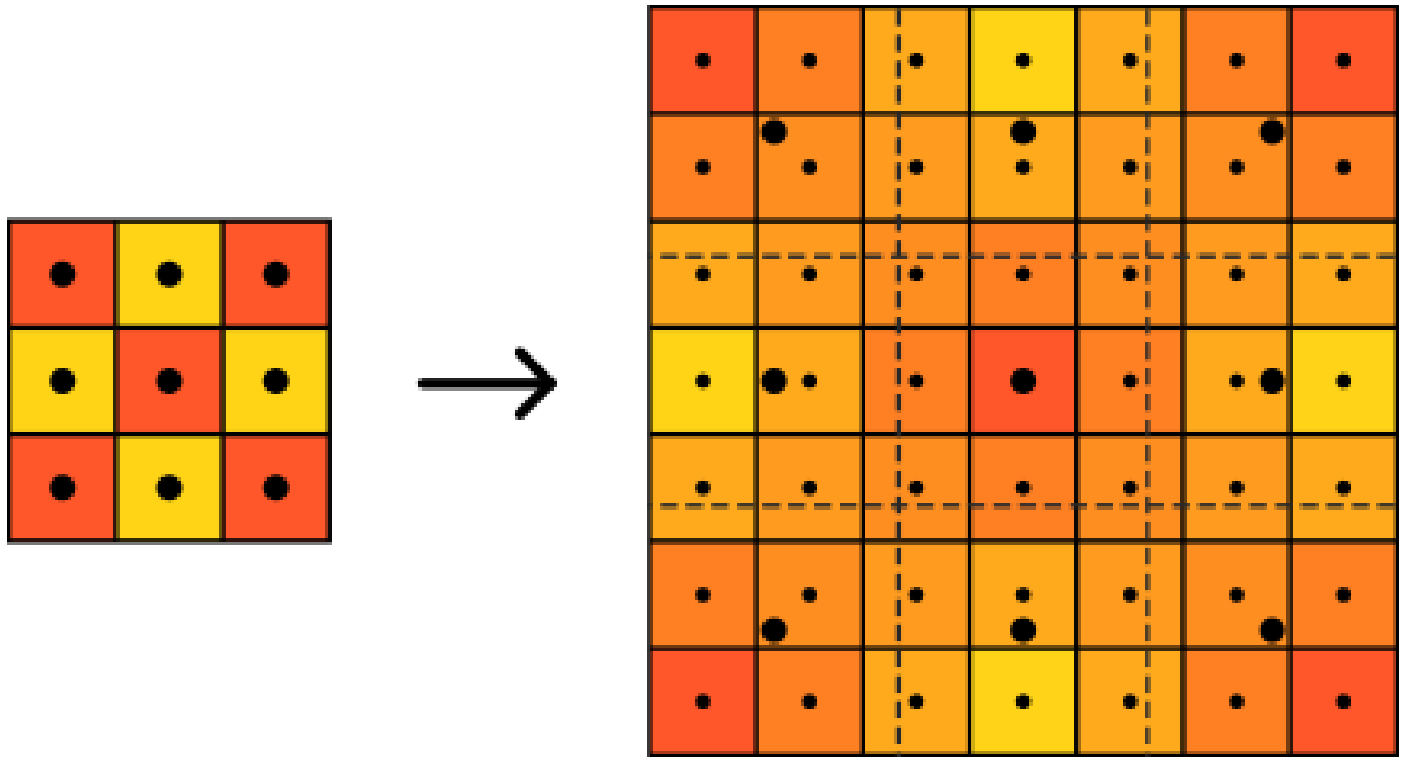
16 gray-levels



8 gray-levels

Resizing (Zooming and Shrinking) Digital Images

Resizing requires two steps:
1) the creation of new pixel locations
2) the assignment of gray levels to those new locations.



Resizing Digital Images

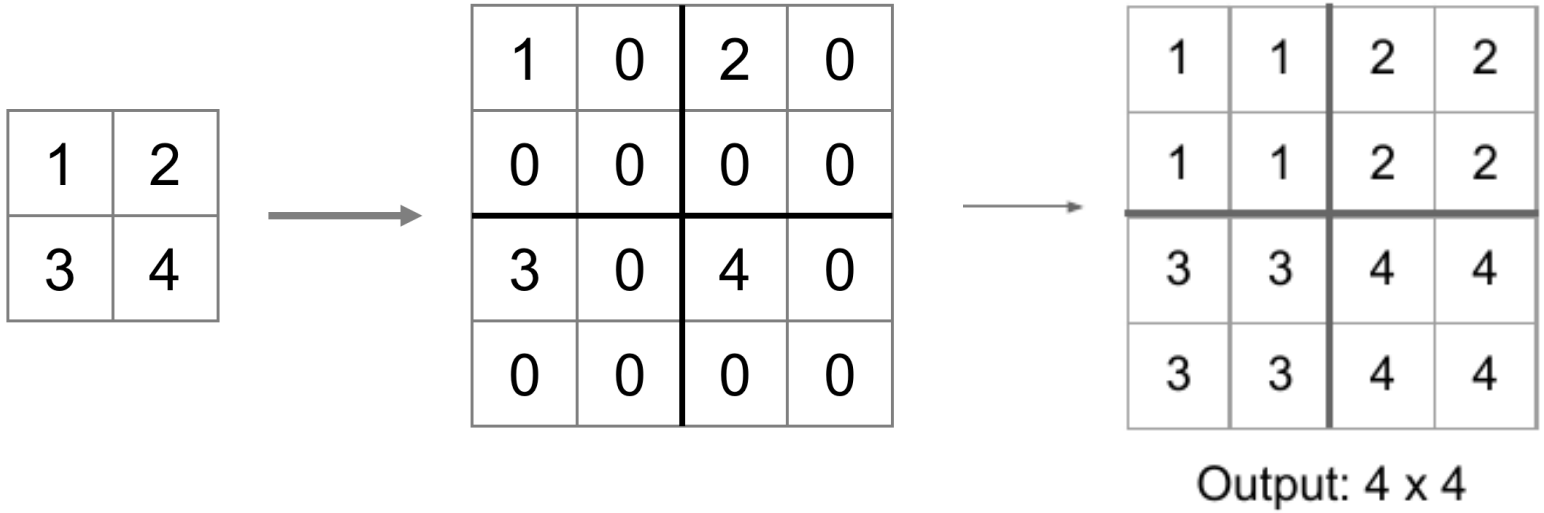
Resizing requires two steps:
1) the creation of new pixel locations
2) the assignment of gray levels to those new locations.

- Suppose that we have an image of size 500×500 pixels and we want to enlarge it 1.5 times to 750×750 pixels.
- We create a 750×750 image and lay it on the original image.
- To assign a gray-level to a pixel in the zoomed image, we look for the closest pixels in the original image.
- We compute the gray-level of the pixel in the zoomed image using the gray-levels of the closest pixels.

The simplest way: ***Nearest neighbor interpolation***

Resizing Digital Images

Nearest-neighbor interpolation; the output pixel is assigned the value of the nearest pixel. No other pixels are considered.



Bilinear interpolation; The output pixel value is a weighted average of pixels in the nearest 2-by-2 neighborhood.

Bicubic interpolation; the output pixel value is a weighted average of pixels in the nearest 4-by-4 neighborhood

Resizing Digital Images



Nearest-neighbor interpolation



Bilinear interpolation



64×64 pixels

1024×1024 pixels

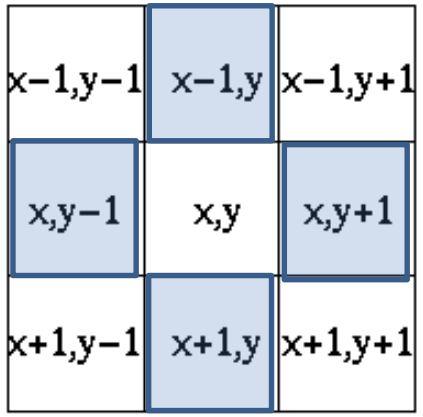
Basic relationships betweeen pixels

- **Neighbors** of a pixel p at coordinates (x, y)

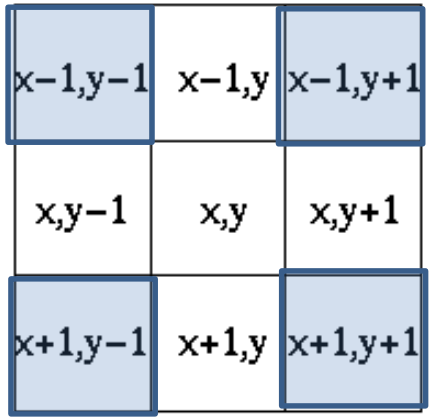
➢ **4-neighbors of $\boldsymbol{p}$**, denoted by $\boldsymbol{N_4(p)}$:
$(x-1, y)$, $(x+1, y)$, $(x, y-1)$, and $(x, y+1)$.

➢ **4 diagonal neighbors of $\boldsymbol{p}$**, denoted by $\boldsymbol{N_D(p)}$:
$(x-1, y-1)$, $(x+1, y+1)$, $(x+1, y-1)$, and $(x-1, y+1)$.

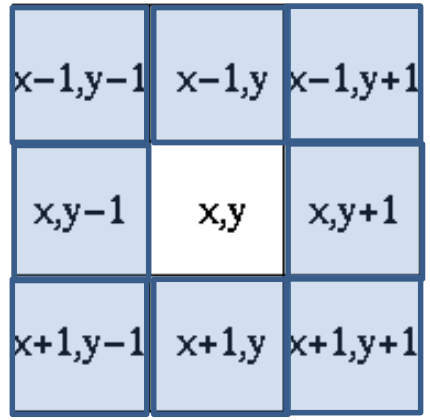
➢ **8 neighbors of $\boldsymbol{p}$**, denoted $\boldsymbol{N_8(p)}$
$N_8(p) = N_4(p)\ U\ ND(p)$



4-neighbors of p



4 diagonal neighbors of p



8 neighbors of p

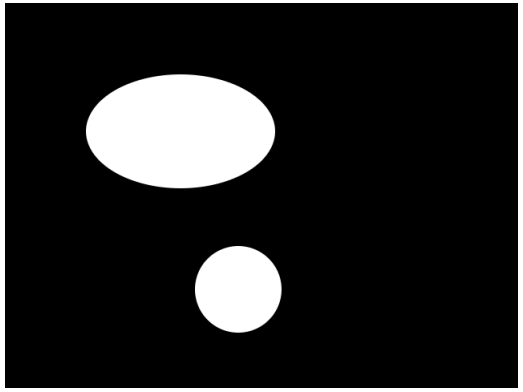
Image Operations on a Pixel Basis

Image multiplication

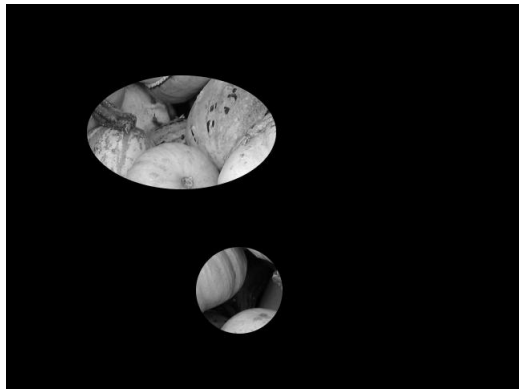
- "Multiplication of one image by another," means that the multiplication is carried out between *corresponding* pixels in the two images (as in array multiplication).
- For example, if f and g are images, the first element of the image formed by "multiplying" f by g is simply the first pixel in f multiplied by the first pixel in g.
- Other arithmetic and logic operations are similarly defined between corresponding pixels in the images involved.



$f(x, y)$



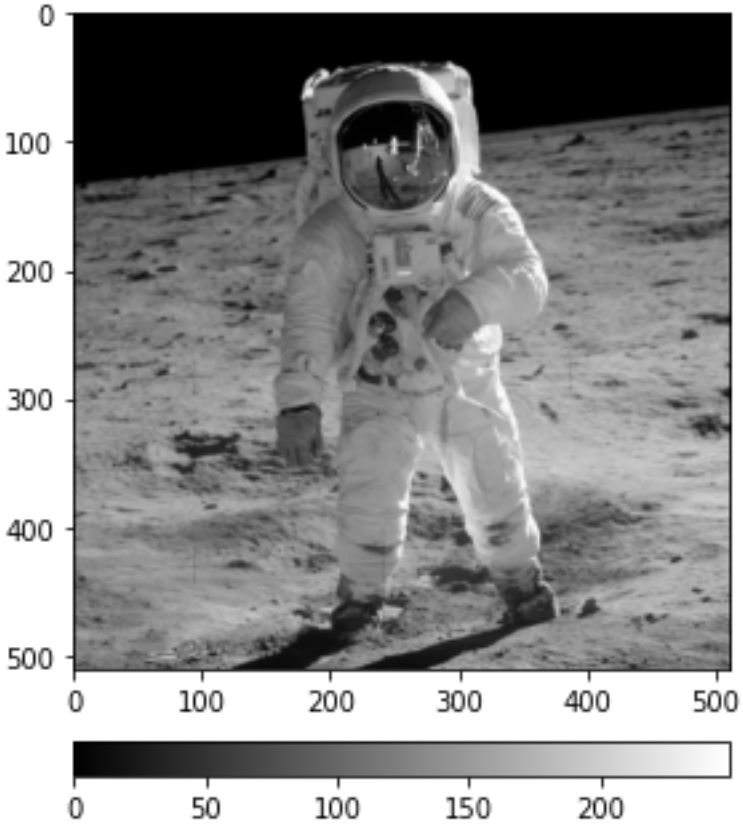
$g(x, y)$



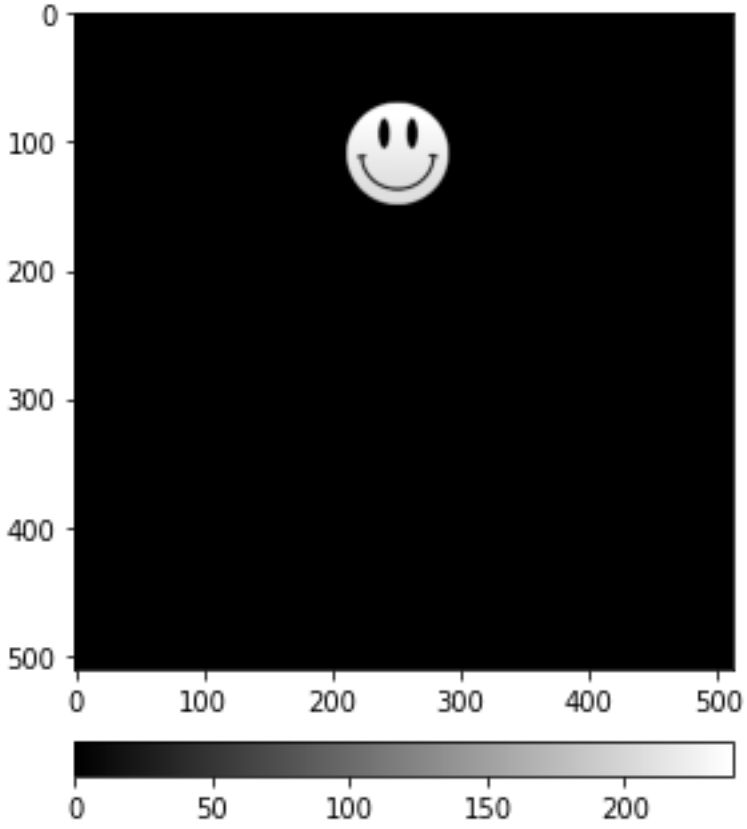
$$h(x, y) = f(x, y)g(x, y)$$

Image Operations on a Pixel Basis

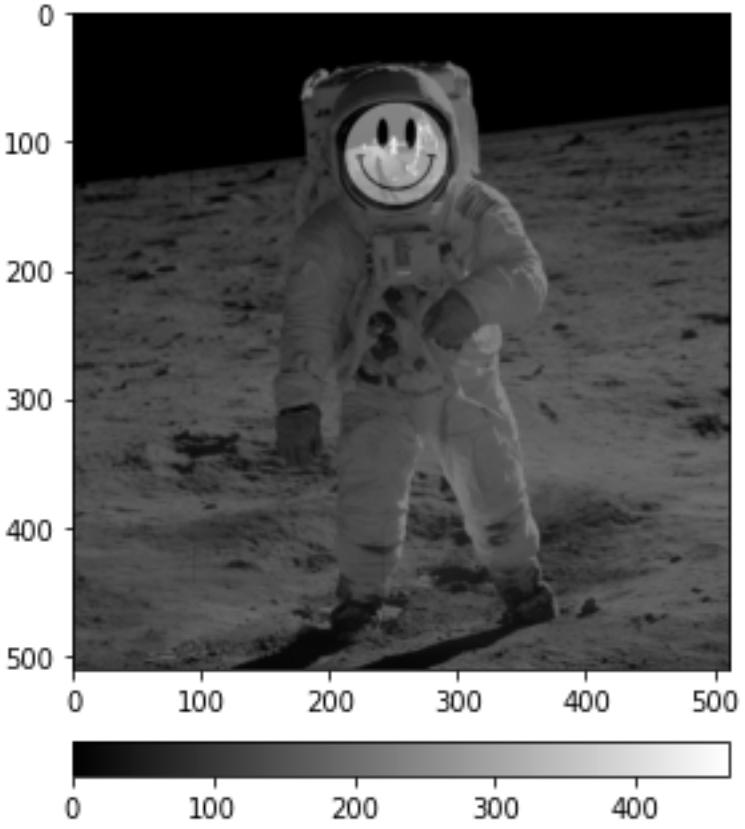
Image addition



$f(x, y)$



$g(x, y)$



$h(x, y) = f(x, y) + g(x, y)$

Image Operations on a Pixel Basis

Image subtraction

Example: Background subtraction is used for detecting moving objects from videos captured with a fixed camera.

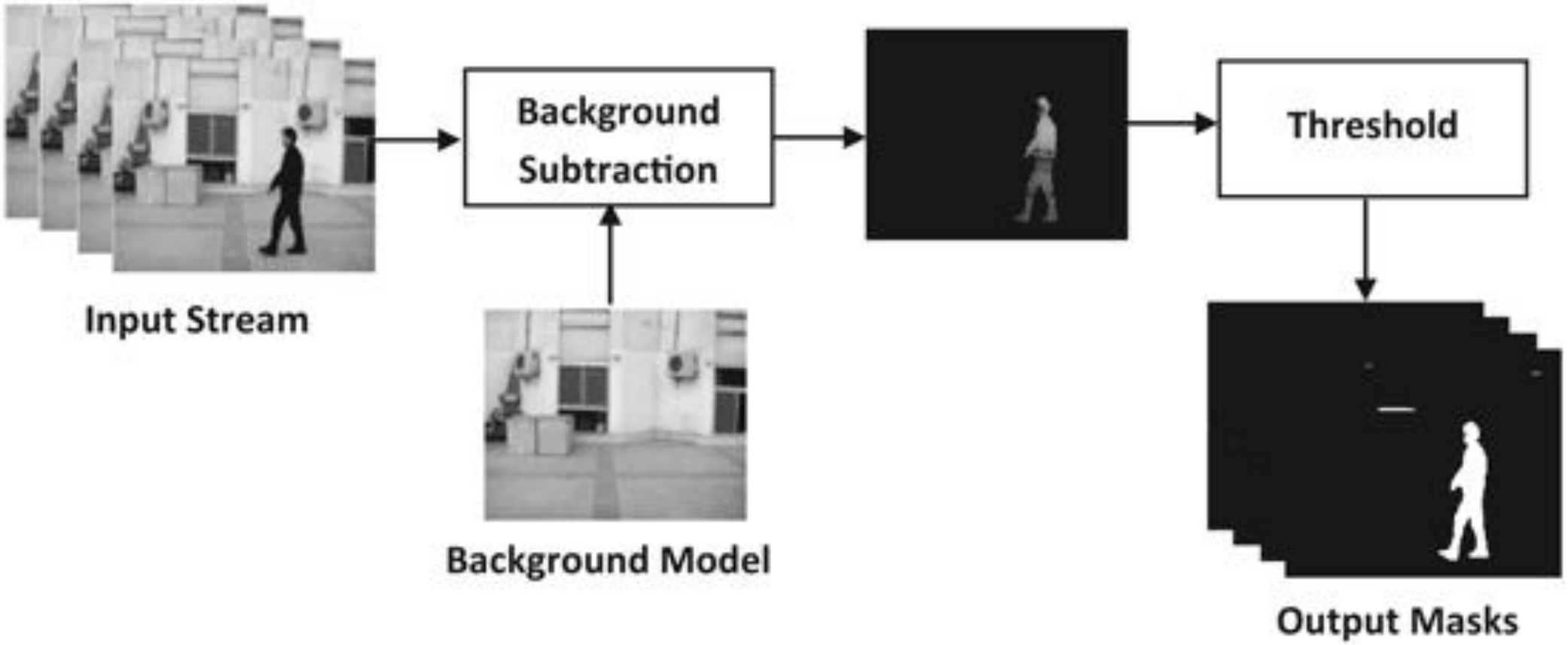
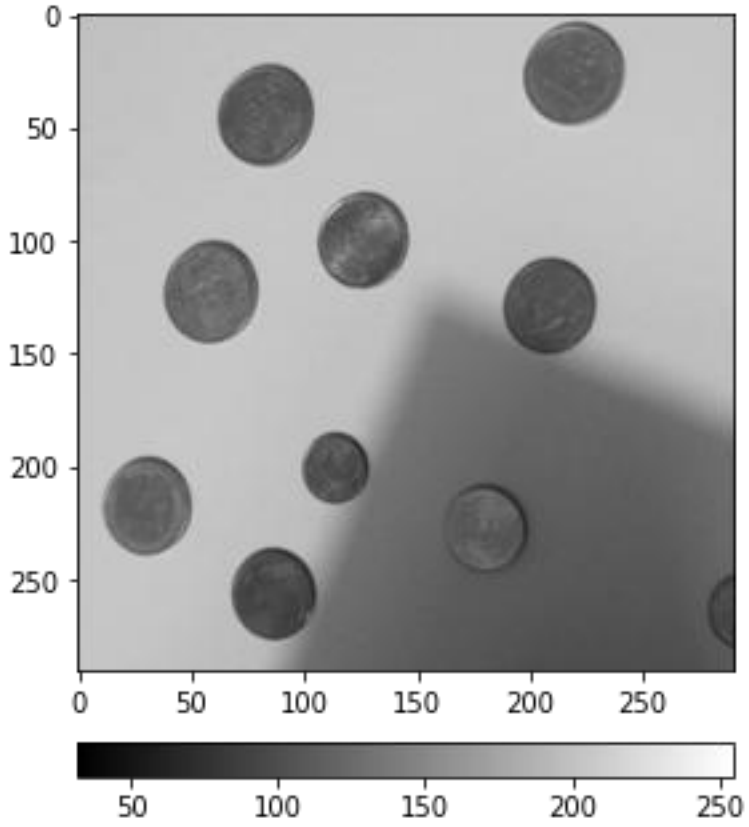
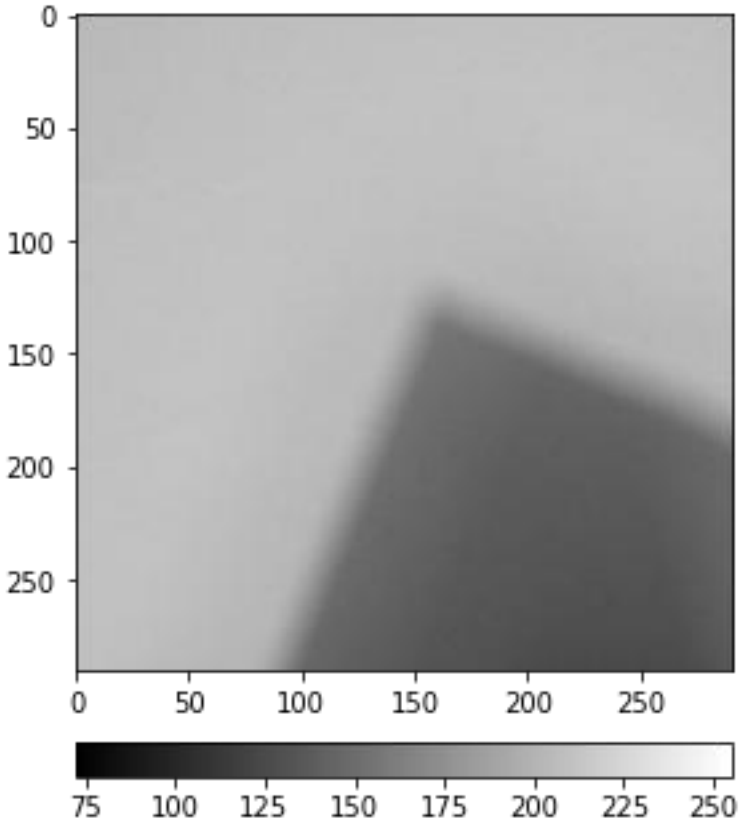


Image Operations on a Pixel Basis

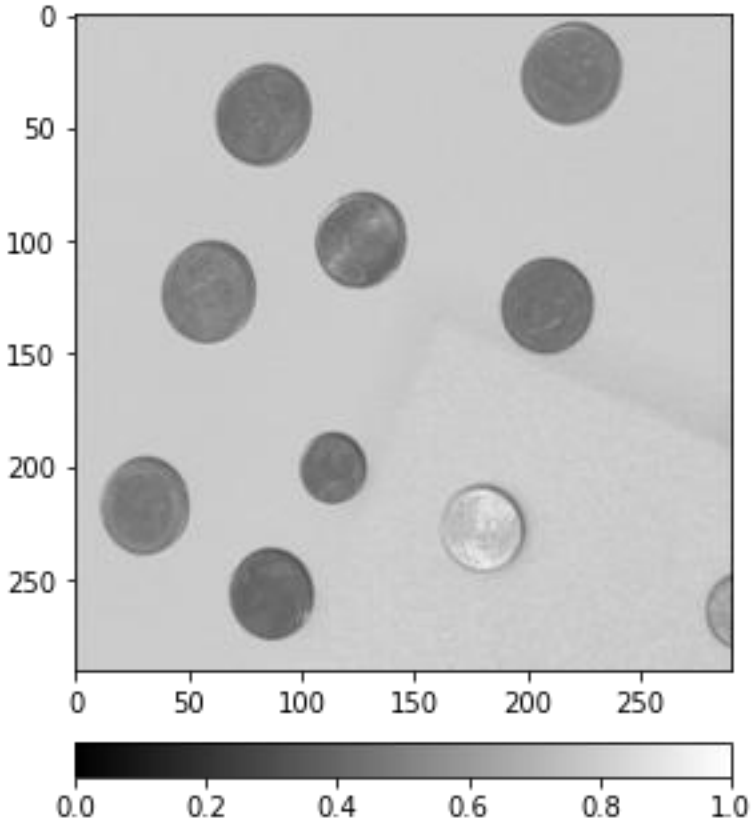
Image division



$f(x,y)$



$g(x,y)$



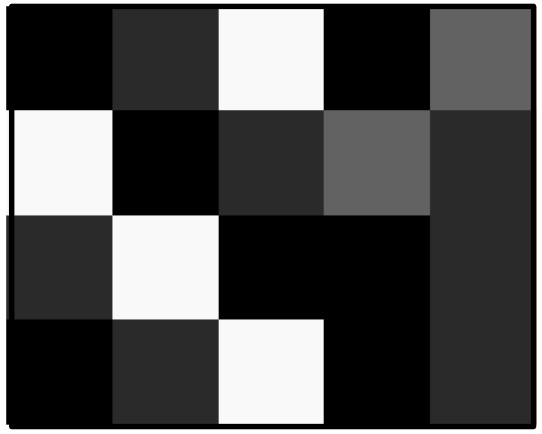
$h(x,y) = f(x,y)/g(x,y)$

Dividing two images allows to remove a non-homogeneous illumination (shading correction)

Image Histogram

- The histogram of a digital image with gray levels in the range $[0, L-1]$ is a discrete function $h(r_k) = n_k$, where r_k is the kth gray level and n_k is the number of pixels in the image having gray level r_k.

Example:



$L = 256$

$h(0) = 7$, $h(43) = 7$, $h(100) = 2$, $h(250) = 4$

$h(r_k) = 0$ for $r_k = 1,2,,\ldots,255$ and $r_k \neq 0,43,100,250$

0	43	250	0	100
250	0	43	100	43
43	250	0	0	43
0	43	250	0	43

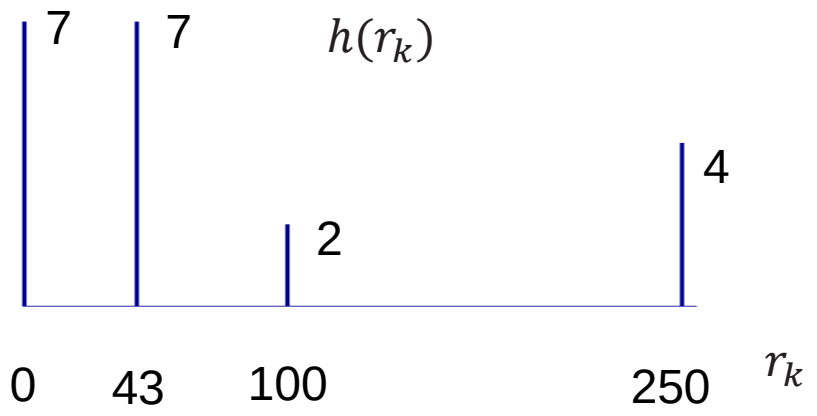
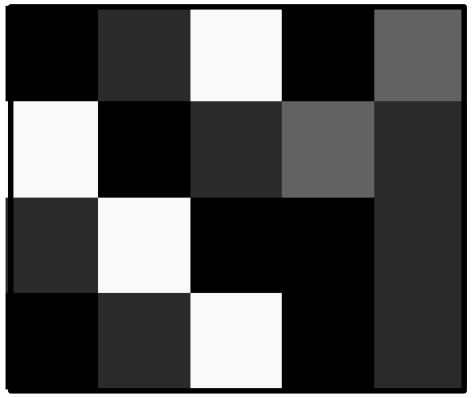


Image Histogram

Normalized Image Histogram: We normalize a histogram by dividing each of its values by the total number of pixels in the image, denoted by n.

- A normalized histogram is given by $p(r_k) = n_k/n$, for $k = 0, 1, \dots, L-1$.
- Loosely speaking, $p(r_k)$ gives an estimate of the probability of occurrence of gray level r_k. Note that the sum of all components of a normalized histogram is equal to 1.

Example:



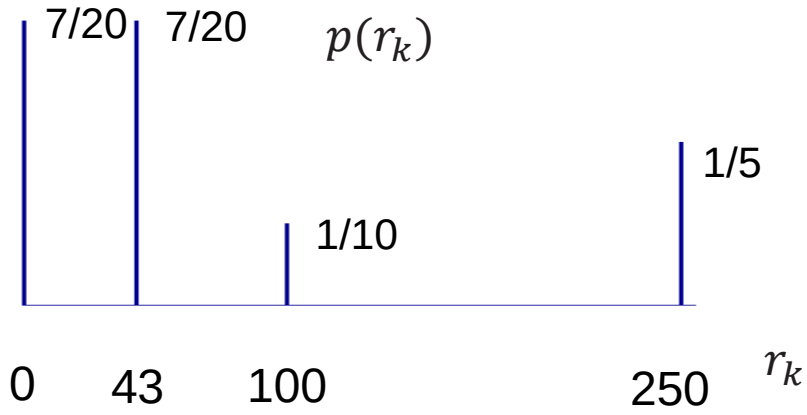
$L = 256$

$n = 4\text{x}5 = 20$

$p(0) = 7/20$, $p(43) = 7/20$, $p(100) = 2/20$, $h(250) = 4/20$

$p(r_k) = 0$ for $r_k = 1,2,,\dots,255$ and $r_k \neq 0,43,100,250$

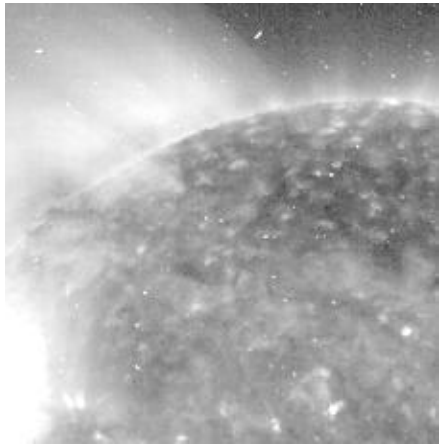
0	43	250	0	100
250	0	43	100	43
43	250	0	0	43
0	43	250	0	43



Images and corresponding histograms



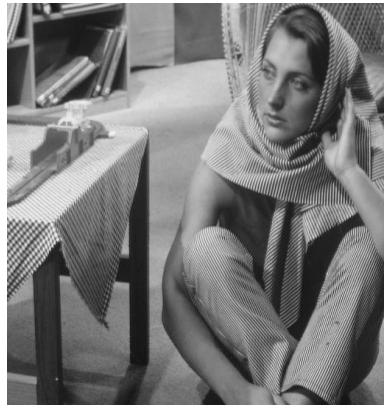
Low-contrast image



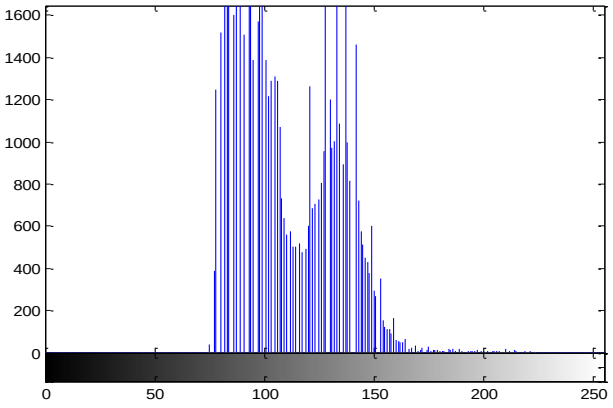
Bright image

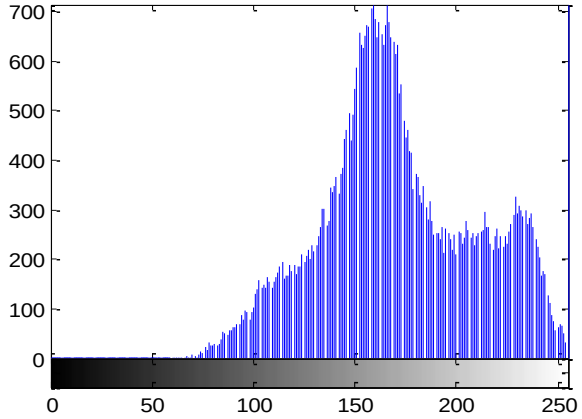


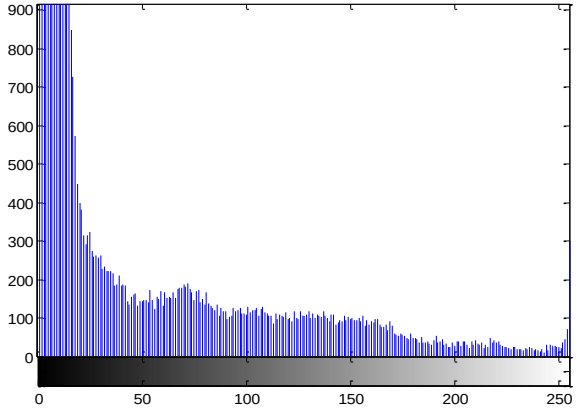
Dark image

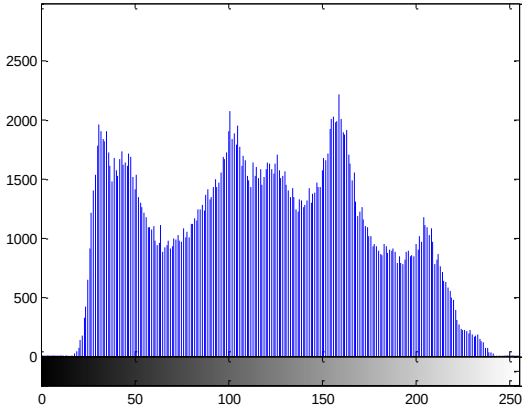


High-contrast image





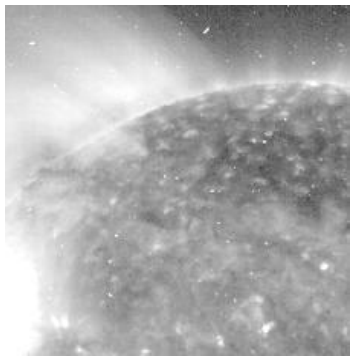




Images and corresponding histograms



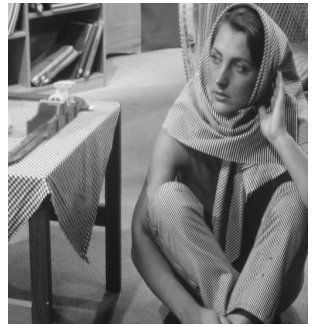
Low-contrast image



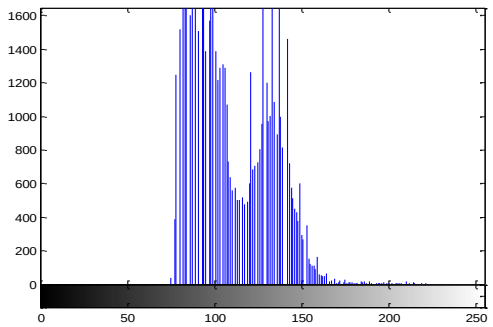
Bright image

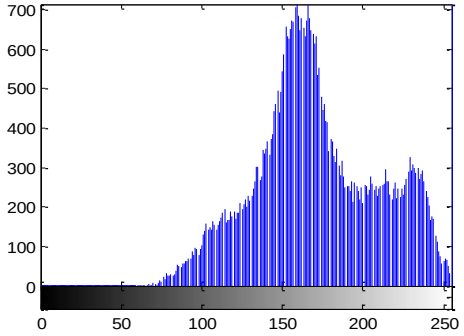


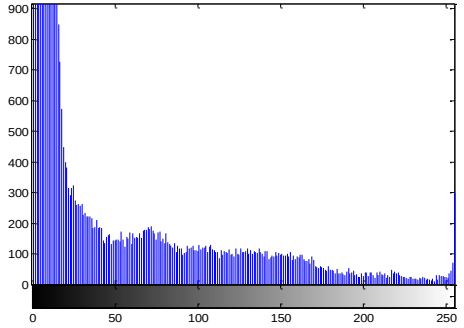
Dark image

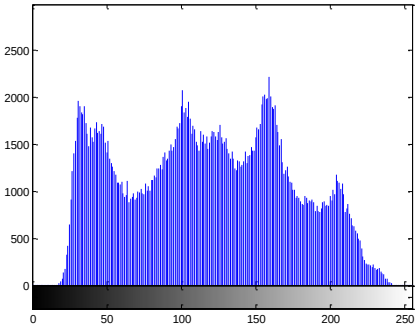


High-contrast image





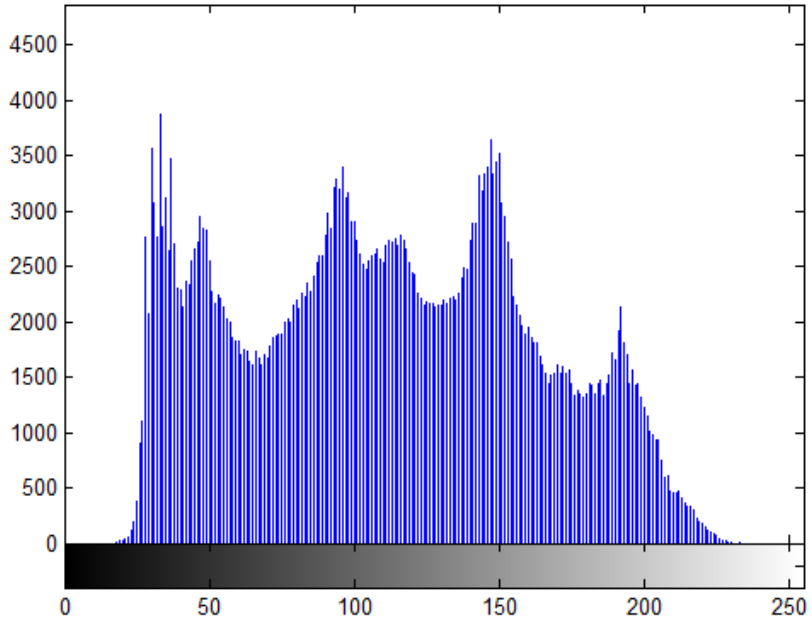




- In the dark image, the components of the histogram are concentrated on the low (dark) side of the gray scale.
- Similarly, the components of the histogram of the bright image are biased toward the high side of the gray scale.
- An image with low contrast has a histogram that is narrow.

Images and corresponding histograms



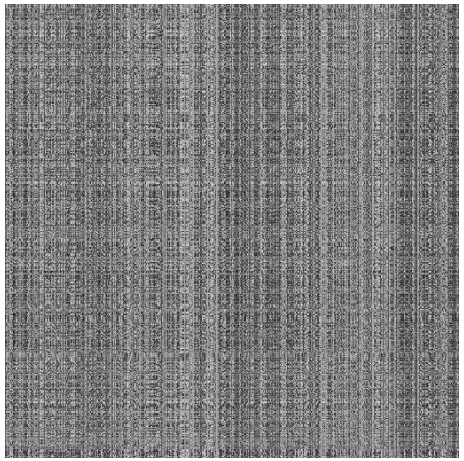


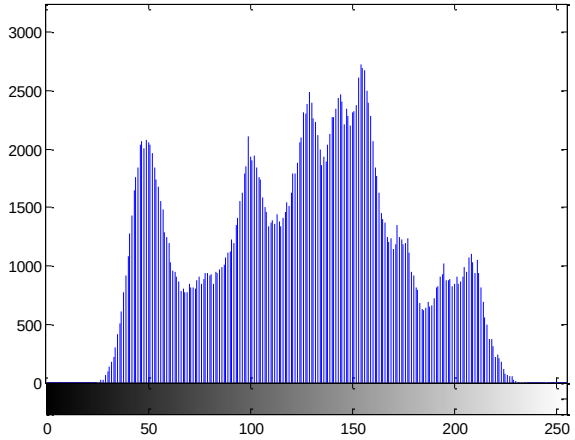
High-contrast images

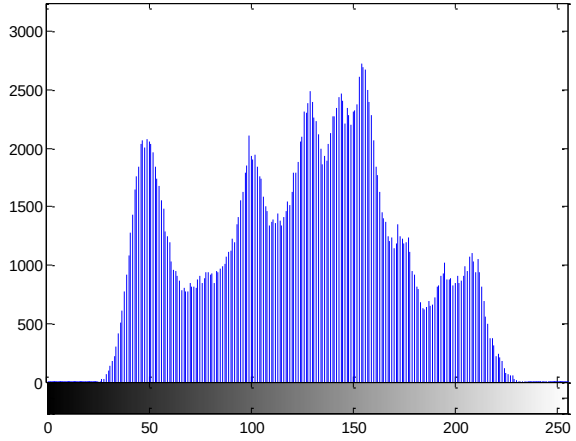
- The components of the histogram in the high-contrast image cover a broad range of the gray scale.
- An image whose pixels tend to occupy the entire range of possible gray levels will have an appearance of high contrast and will exhibit a large variety of gray tones.
- A high-contrast image shows a great deal of gray-level detail and has high dynamic range.

The two images below have the same histogram.
The image on the right is obtained by shuffling the pixels of the image on the left.









A histogram does *not* give information about the spatial organization of the pixels. It does not reveal the structures and/or patterns present within the associated image.
It only gives «the distribution» of pixel values.

Spatial Transformations

Spatial domain methods are procedures that operate directly on the pixels of an image (or a set of images).

$$g(x,y) = T[f(x,y)]$$

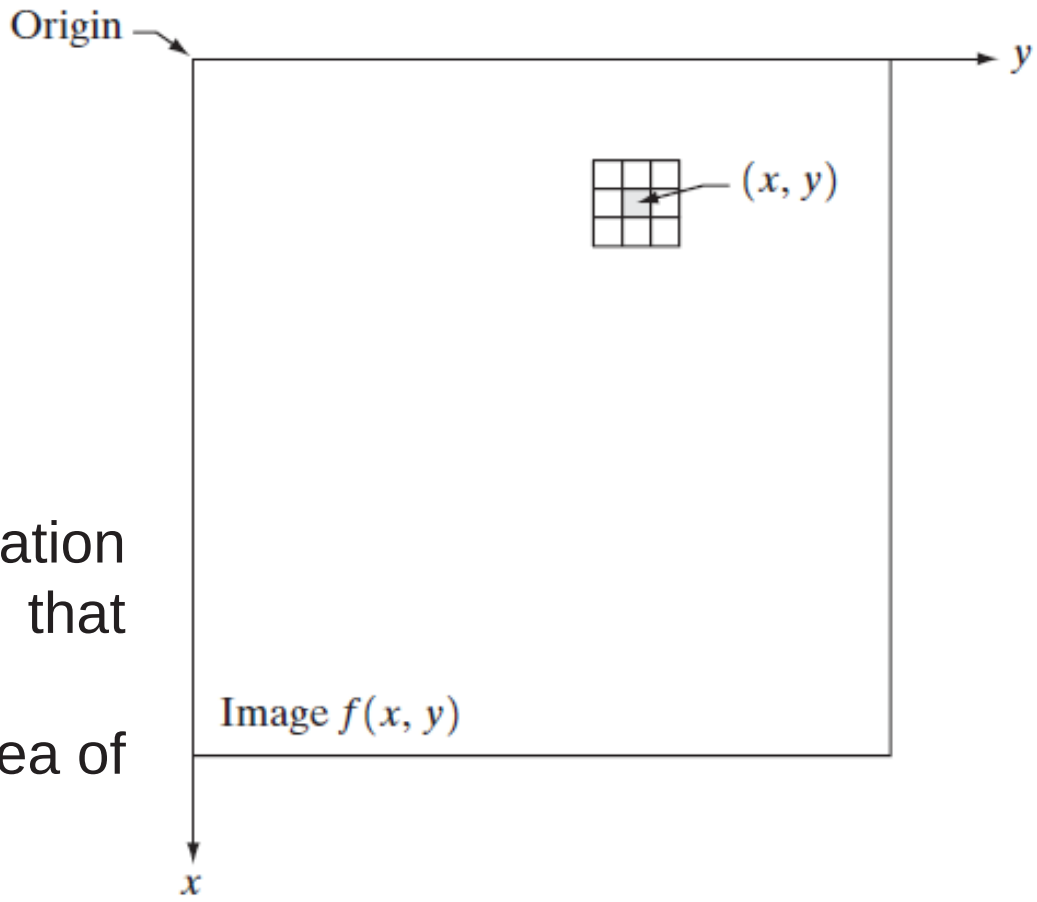
$f(x,y)$: input image

$g(x,y)$: output image

T : an operator on f defined over

a neighborhood of point (x,y)

- The operator T is applied at each location (x,y) to yield the output image g, at that location.
- The process utilizes only the pixels in the area of the image spanned by the neighborhood.



Gray Level (Intensity) Transformations

$$g(x, y) = T[f(x, y)]$$

- The simplest form of *T* is when the neighborhood is of size 1×1 (that is, a single pixel).
- In this case, *g* depends only on the value of *f* at (x, y)
- *T* becomes a *gray-level* (also called an *intensity* or *mapping*) *transformation function* of the form:

$$s = T(r)$$

r and s are variables denoting, respectively, the gray level of $f(x, y)$ and $g(x, y)$ at any point (x, y).
T is a transformation that maps a pixel value r into a pixel value s.

Techniques in this category often are referred to as *point processing*.

Gray Level (Intensity) Transformations

$$g(x, y) = T[f(x, y)]$$

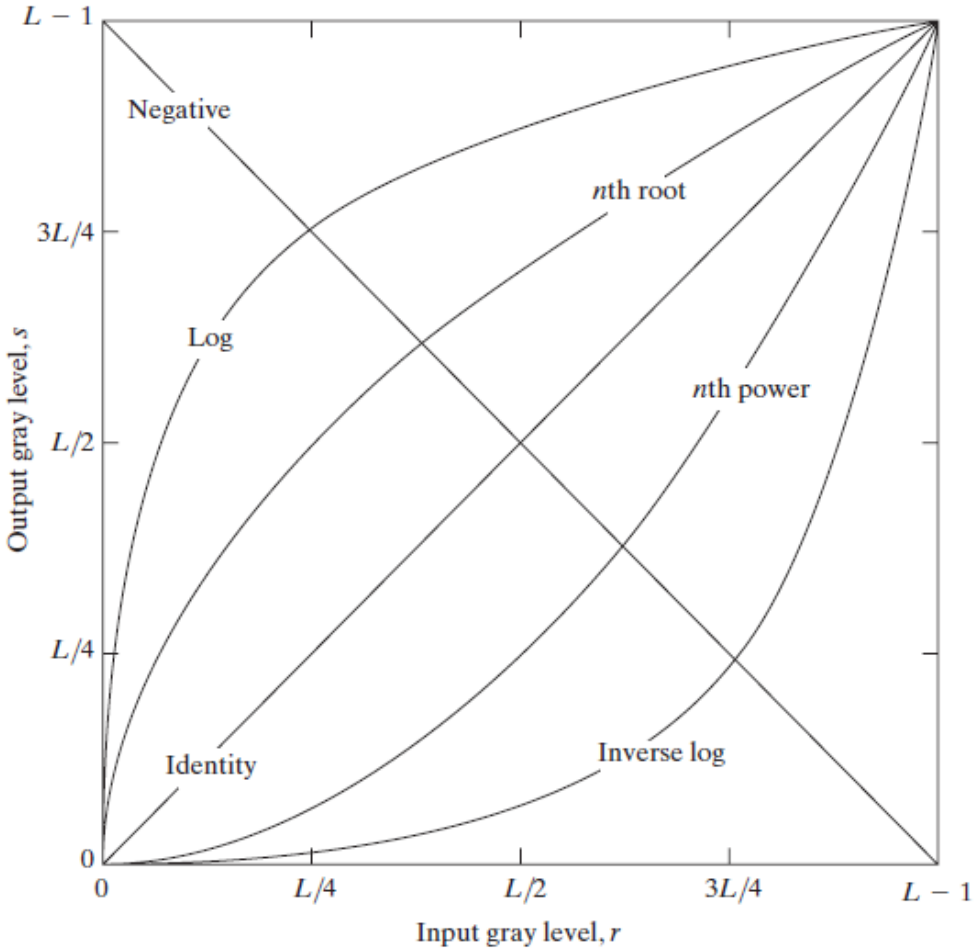
$$s = T(r)$$

r and s are the gray level values of $f(x, y)$ and $g(x, y)$ at any point (x, y).
T is a transformation that maps a pixel value r into a pixel value s.



Gray Level (Intensity) Transformations

Basic gray-level transformation functions used for image enhancement.

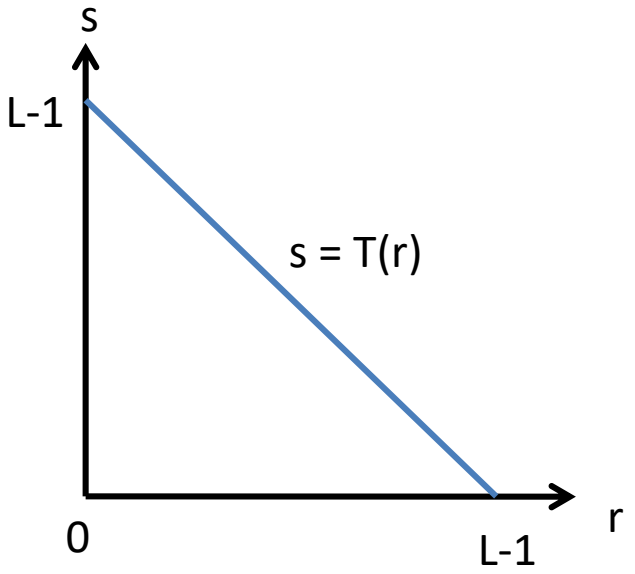


$$s = T(r)$$

- Linear transformations (negative and identity transformations)
- Logarithmic transformations (log and inverse-log transformations)
- Power-law transformtaions (*n*th power and *n*th root transformations)

Gray Level (Intensity) Transformations

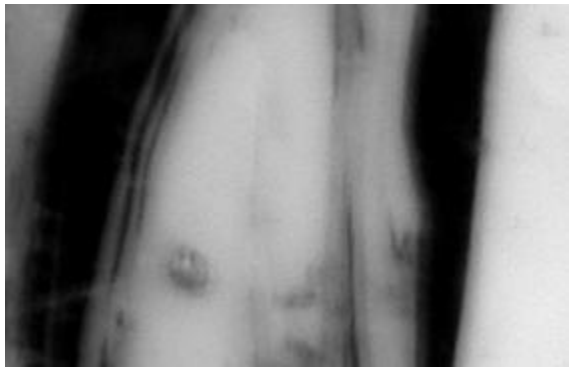
Image Negatives



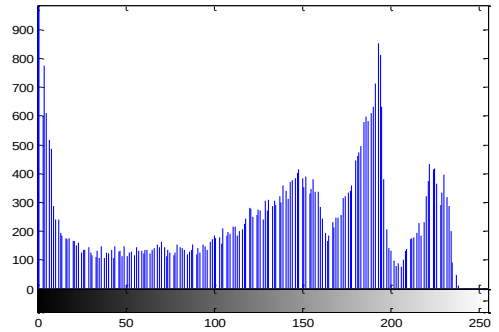
The negative of an image with gray levels in the range $[0, L-1]$ is obtained by using the negative transformation:

$$s = T(r) = L - 1 - r$$

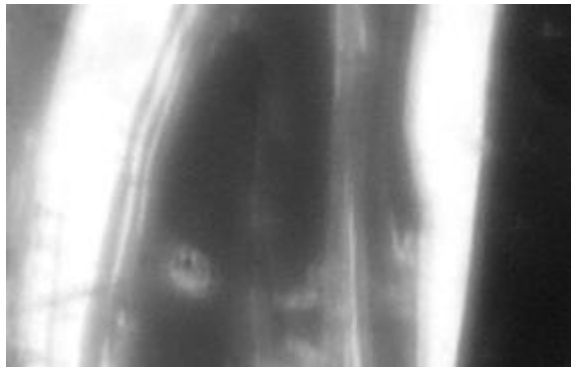
$$g(x, y) = T[f(x, y)]$$



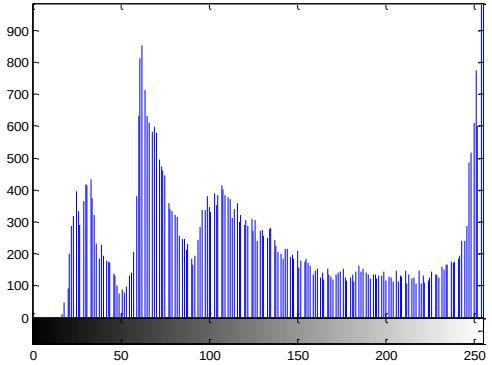
Input image $f(x, y)$



Histogram of the input image



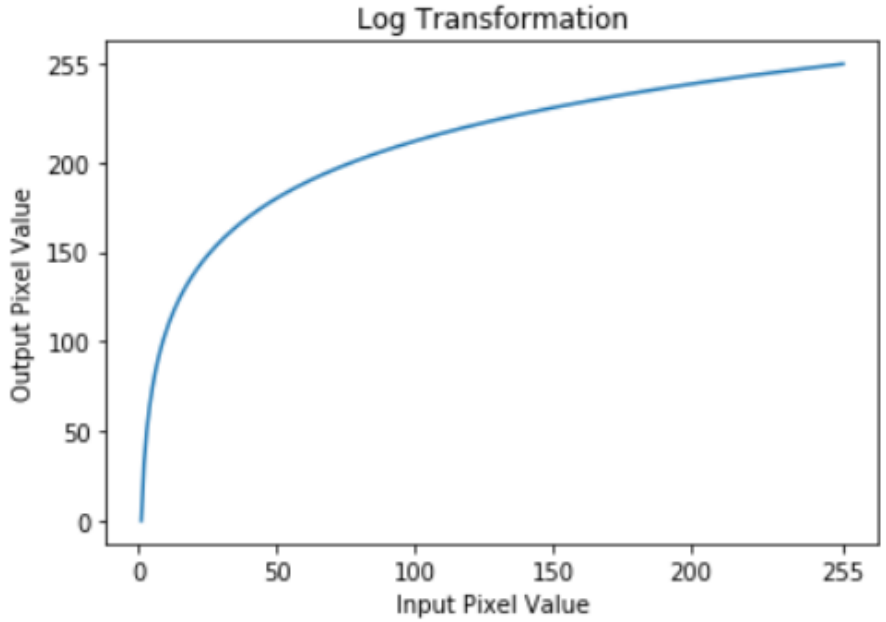
Output image g(x, y)



Histogram of the output image

Gray Level (Intensity) Transformations

Logarithmic Transformation



$$s = T(r)$$

Log Transformations

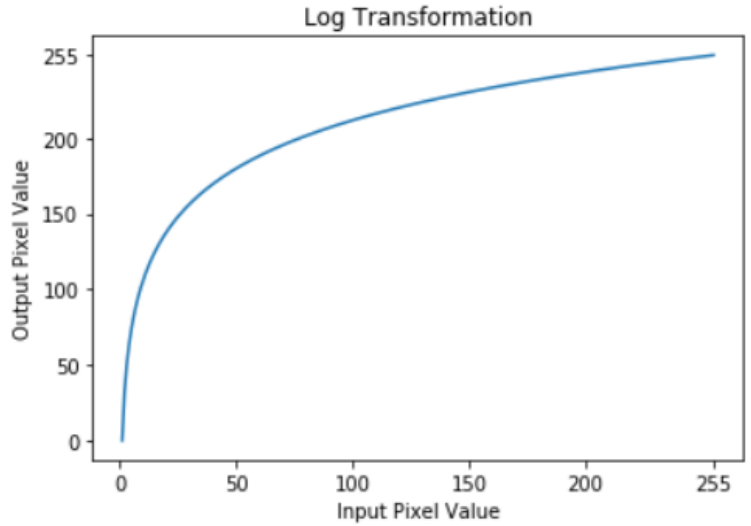
$$s = c\log(1+r)$$

where c is a constant, and $r \geq 0$.

- The shape of the log curve shows that this transformation maps a narrow range of low gray-level values in the input image into a wider range of output levels.
- A wide range of high gray-level values in the input are mapped to a narrow range of output levels.
- We use a transformation of this type to expand the values of dark pixels in an image while compressing the bright pixels.

Gray Level (Intensity) Transformations

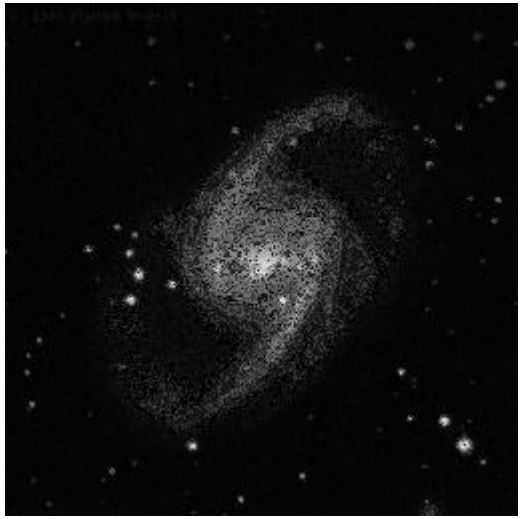
Logarithmic Transformation



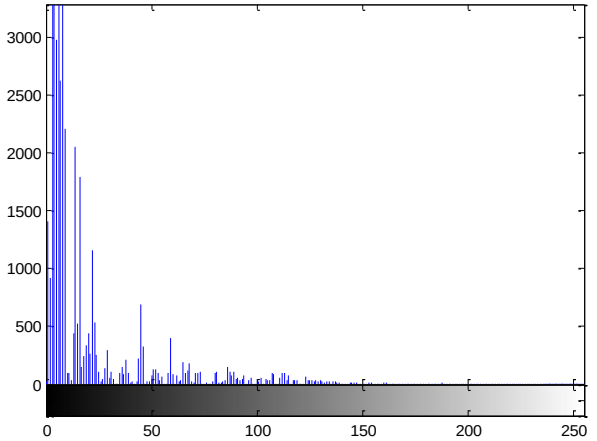
$$s = T(r) = c\log(1 + r)$$

$$g(x, y) = T[f(x, y)]$$

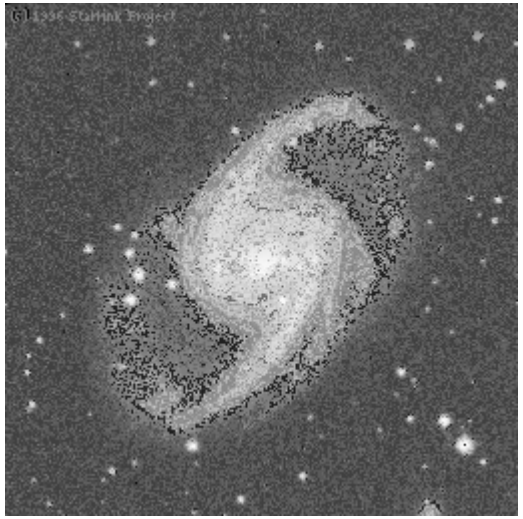
- Dark pixels are expanded to a wider range. More details become visible in dark areas.
- Bright pixels are compressed to a narrower range.



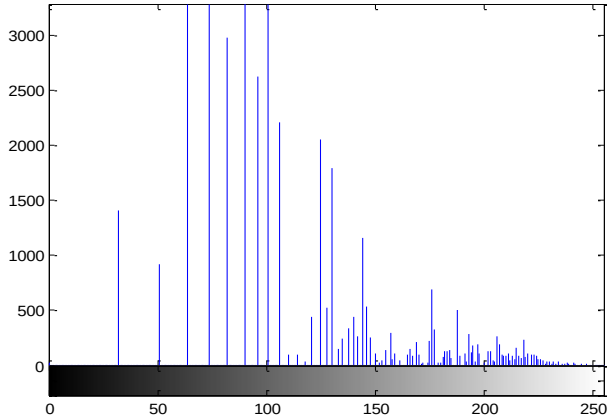
Input image $f(x, y)$



Histogram of the input image



Output image $g(x, y)$

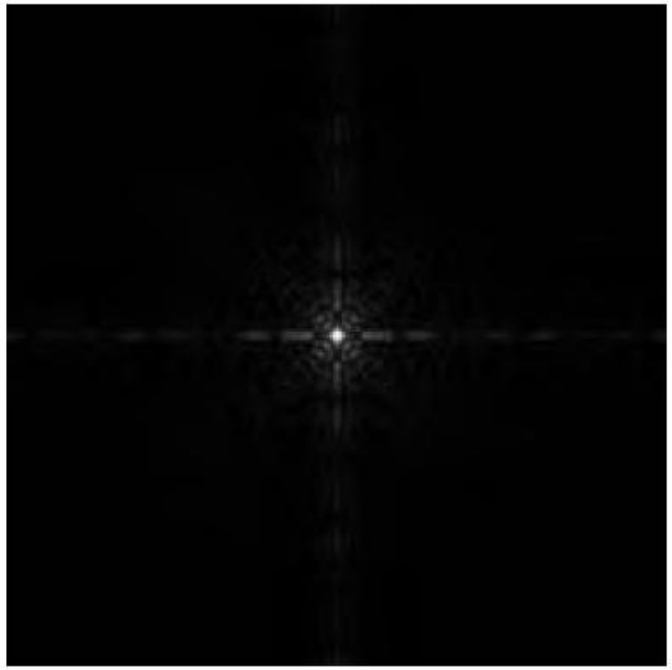


Histogram of the output image

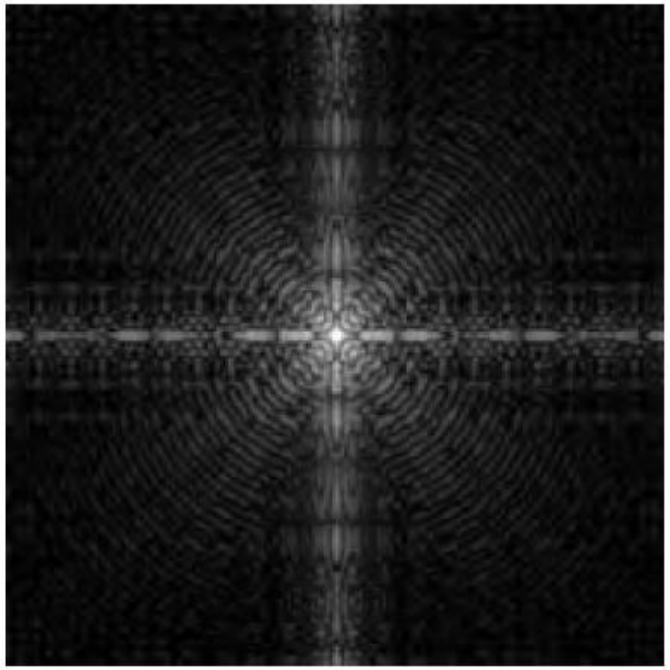
Gray Level (Intensity) Transformations

Logarithmic Transformation

- The log function compresses the dynamic range of images with large variations in pixel values.
- A classic application is to display a 2D Fourier spectrum in which pixel values have a large dynamic range.
- Very large values for low frequency components make it difficult to visualize high frequency components.
- By applying log transformation high frequency components become visible.



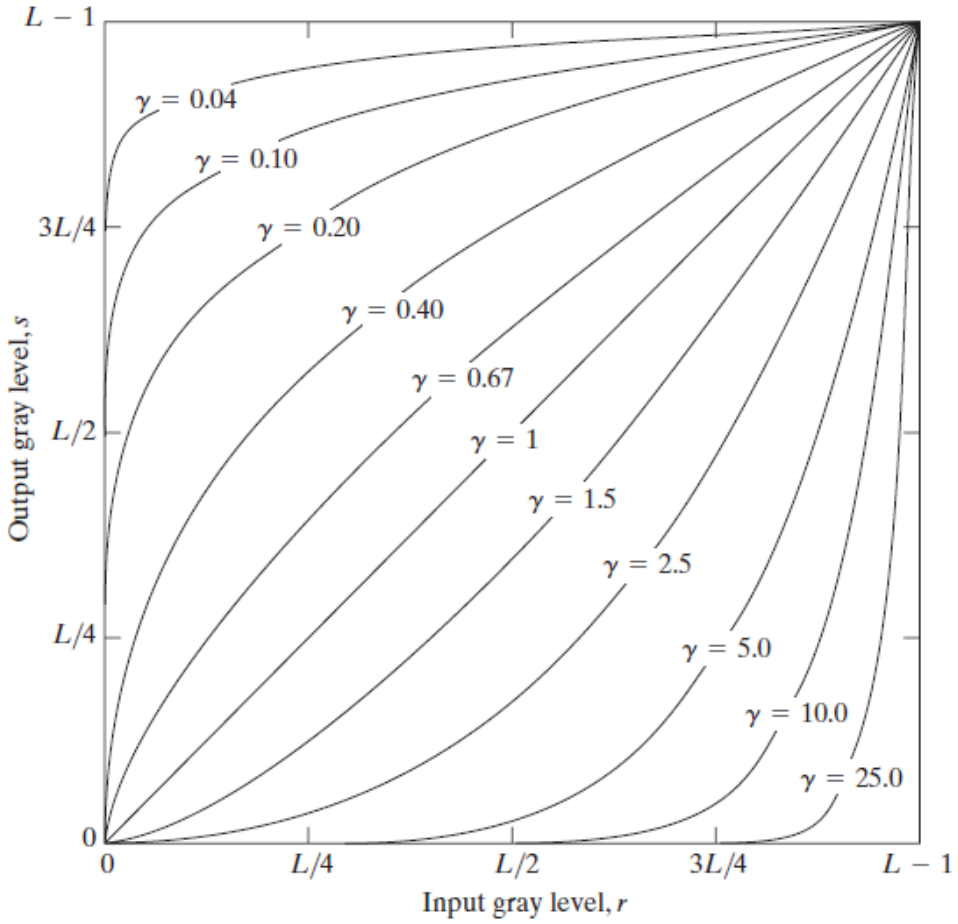
2D Fourier spectrum



Result of applying the log transformation

Gray Level (Intensity) Transformations

Power-law Transformation (Gamma transformation)



Power-law transformations have the basic form

$$s = T(r) = c\, r^{\gamma}$$

where c and γ are positive constants.

- Power-law curves with $0 < \gamma < 1$ map a narrow range of dark input values into a wider range of output values.
- Power-law curves with $\gamma > 1$ map a narrow range of bright input values into a wider range of output values. They compress the range of dark pixels.

Gray Level (Intensity) Transformations

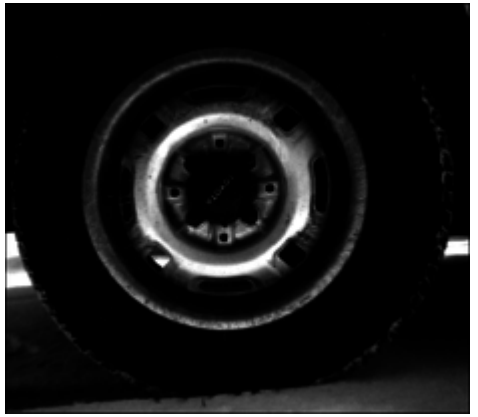
Power-law Transformation (Gamma transformation)

$$s = T(r) = c\, r^{\gamma}$$

Input image

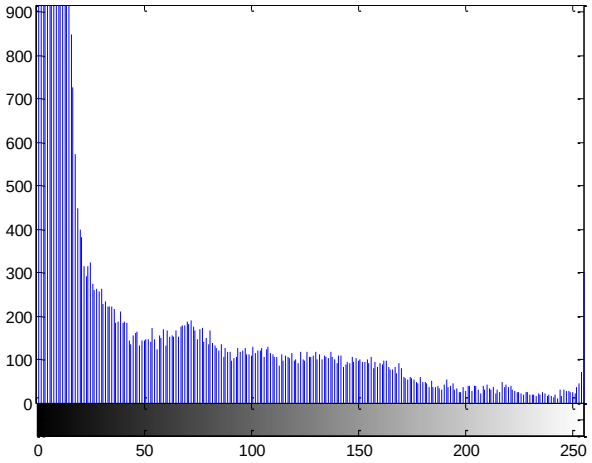


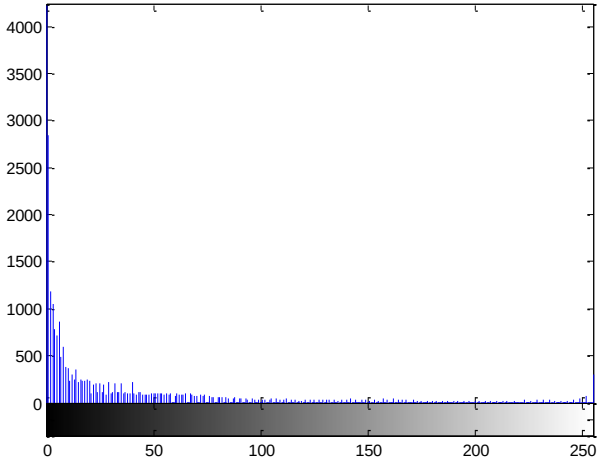
$\gamma = 3$

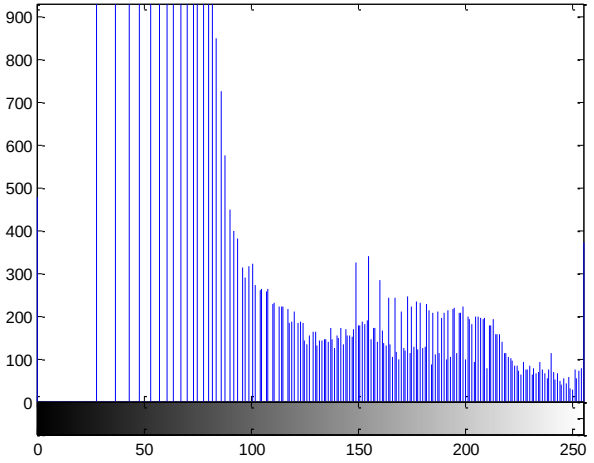


$\gamma = 0.4$









Gray Level (Intensity) Transformations
Power-law Transformation (Gamma transformation)

- The given image is predominantly dark.
- An expansion of dark gray levels are desirable.
- This can be done with a power-law transformation with a fractional exponent ($\mathbf{0} < \boldsymbol{\gamma} < \mathbf{1}$) .
- As gamma is decreased from 0.6 to 0.4, more detail became visible.
- A further decrease of gamma to 0.3 result in reduced contrast to the point where the image started to have a very slight "washed-out" look, especially in the background.



a b
c d

(a) Magnetic resonance (MR) image of a fractured human spine.
(b)–(d) Results of applying gamma transformation with $c = 1$ and $\gamma = 0.6, 0.4, 0.3$, respectively.

Gray Level (Intensity) Transformations

Power-law Transformation (Gamma transformation)

- The image to be enhanced has a washed-out appearance.
- A compression of mid-level gray levels to a darker range is desirable.
- This can be done with a power-law transformation with $\boldsymbol{\gamma > 1}$.
- With gamma values of 3.0 and 4.0, we get images of higher contrast.
- The result obtained with $\gamma = 5$ has areas that are too dark, in which some detail is lost.



a b
c d

(a) Aerial image (b)–(d) Results of applying gamma transformation with $c = 1$ and $\gamma = 3, 4, 5$, respectively.

MATLAB SESSION

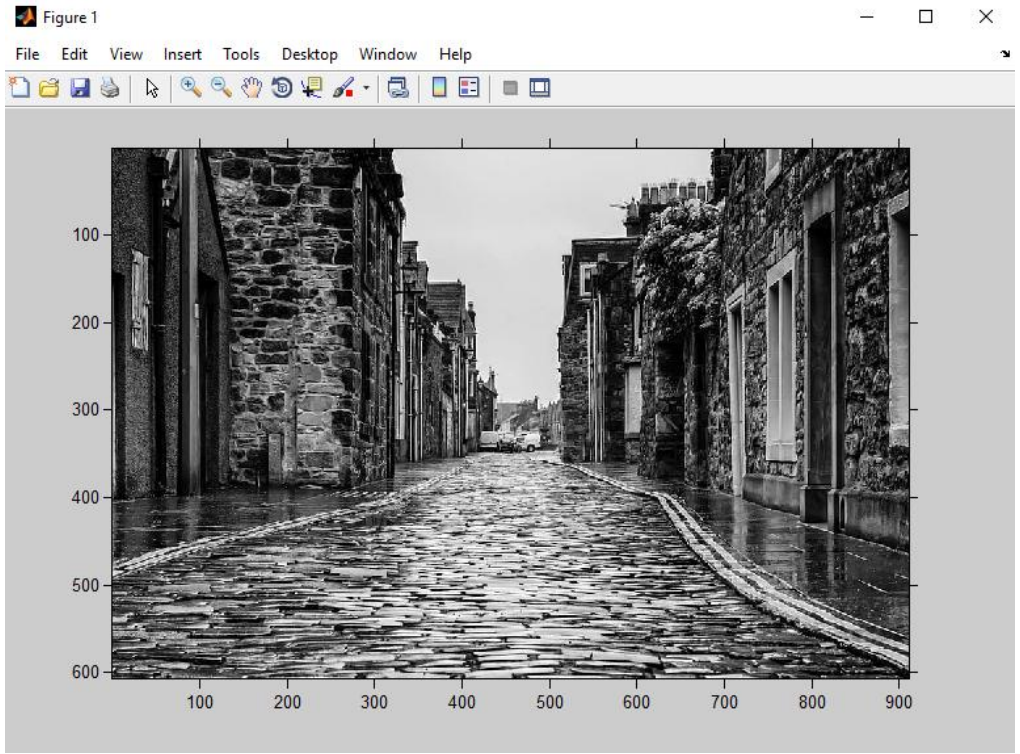
```
Img = imread('road.jpg');
class(Img)

% uint8 -- 8-bit unsigned integer array

Img = rgb2gray(Img);
figure(1);
imshow(Img)
axis on;
size(Img)

Img_s = imresize(Img,0.5,'nearest');
size(Img_s)

%     imresize(A, SCALE, METHOD) returns an image that is SCALE times the
%     size of A, which is a grayscale, RGB, or binary image
%    METHOD can be a string naming a general interpolation method:
%        'nearest'    - nearest-neighbor interpolation
%        'bilinear'   - bilinear interpolation
%        'bicubic'    - cubic interpolation; the default method
```

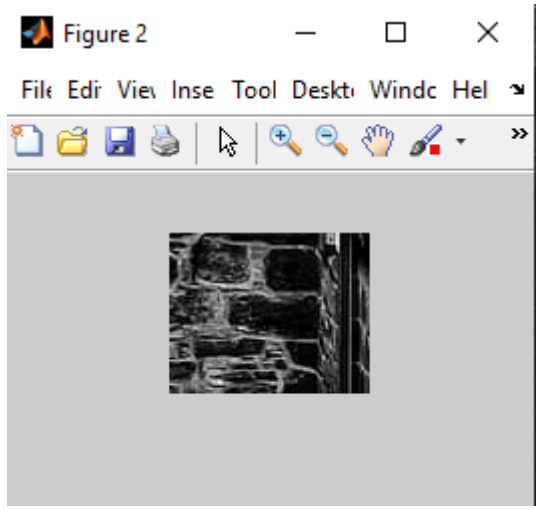


```
P = Img(24,308)  % Graylevel value at location (24,308)

% Extract the region from row 41 to 120 and column 201 to 300
B = Img(41:120,201:300);

figure(2)
imshow(B)

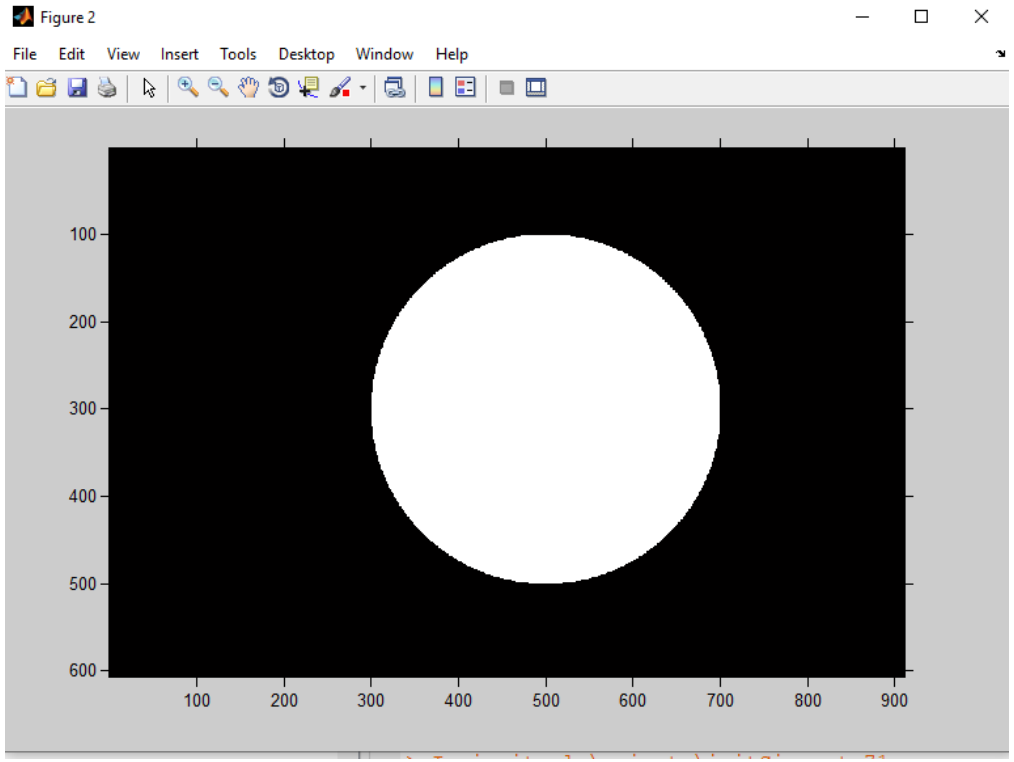
Max_value = max(Img(:))  % Maximum value of Img
Min_value = min(Img(:))  % Minimum value of Img
```



```
% CREATE AN IMAGE OF THE SAME SIZE WITH Img
% WITH A WHITE CIRCLE OF RADIUS 200 PIXELS
% CENTERED AT LOCATION 300 and 500
s = size(Img);
M = s(1);
N = s(2);
C = zeros(M,N);
R_sqr = 200^2;
for i = 1:M
    for j = 1:N
        dist = (300-i)^2 + (500-j)^2;
        if dist < R_sqr
            C(i,j) = 1;
        end
    end
end

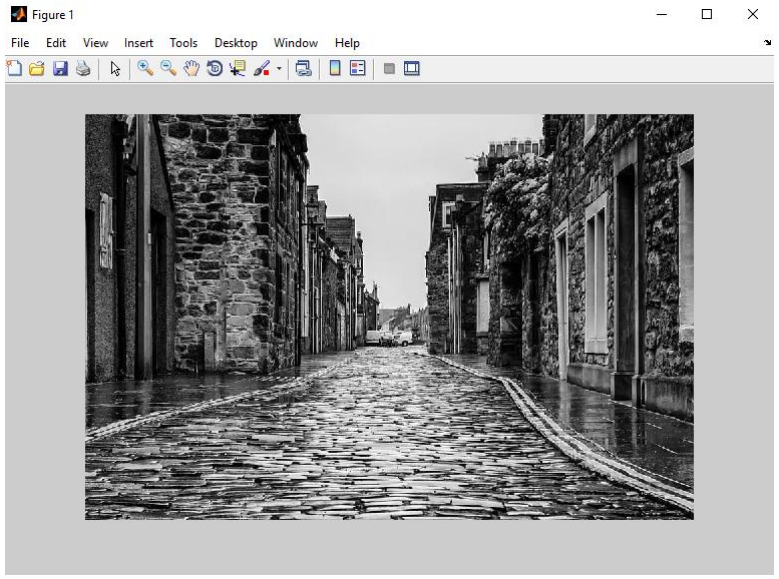
class(C)
% double  -- Double precision floating point number array

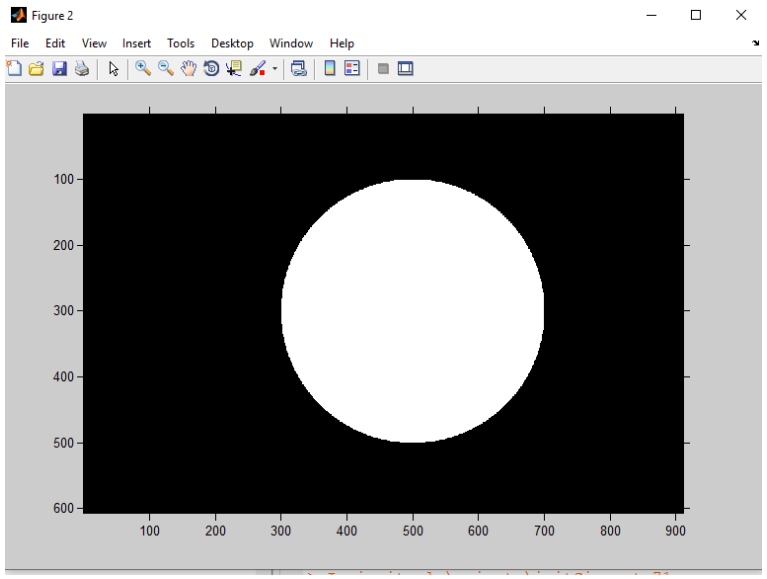
figure(2);
imshow(C);
axis on;
```

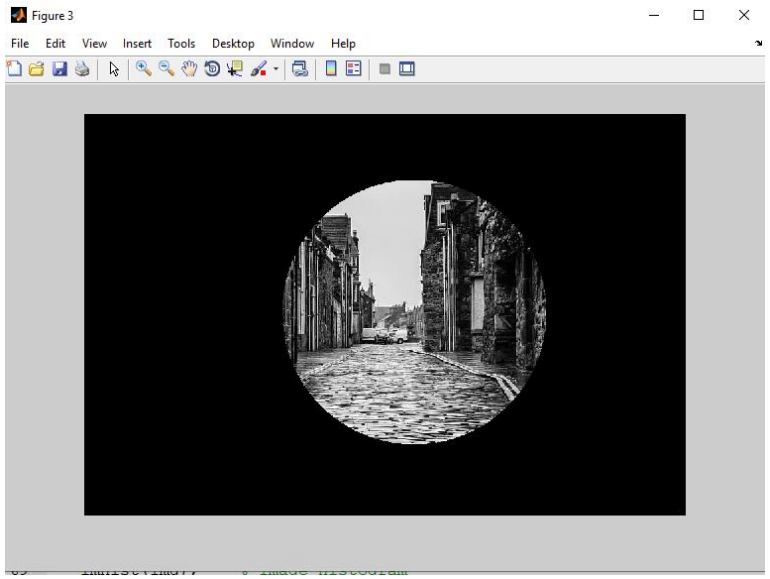


```
Img_C = double(Img).*C;    % Array multiplication, Multiply pixel by pixel
Img_C = uint8(Img_C);

figure(3);
imshow(Img_C);
```

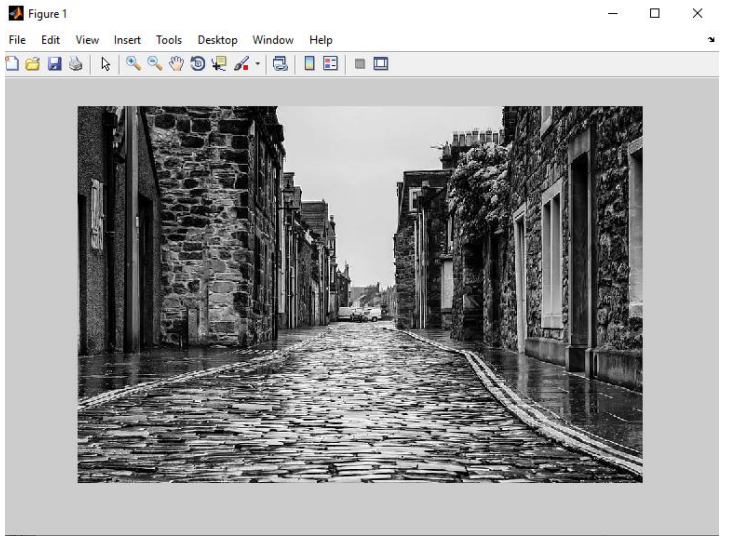


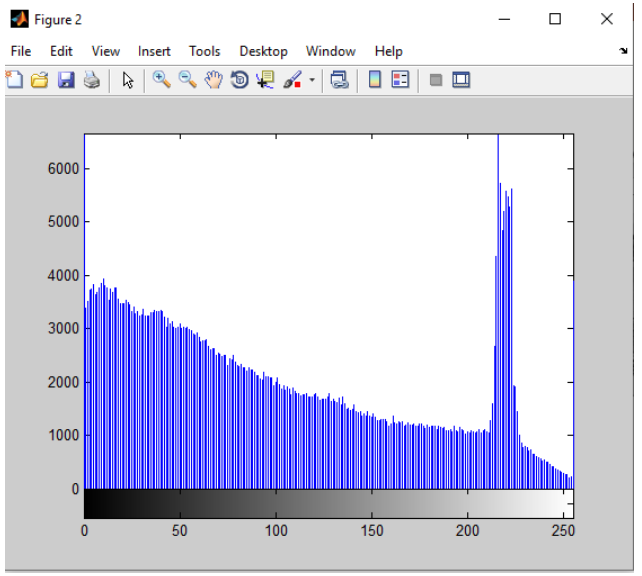




```
close(2); close(3);

figure(2);
imhist(Img);      % Image Histogram
```

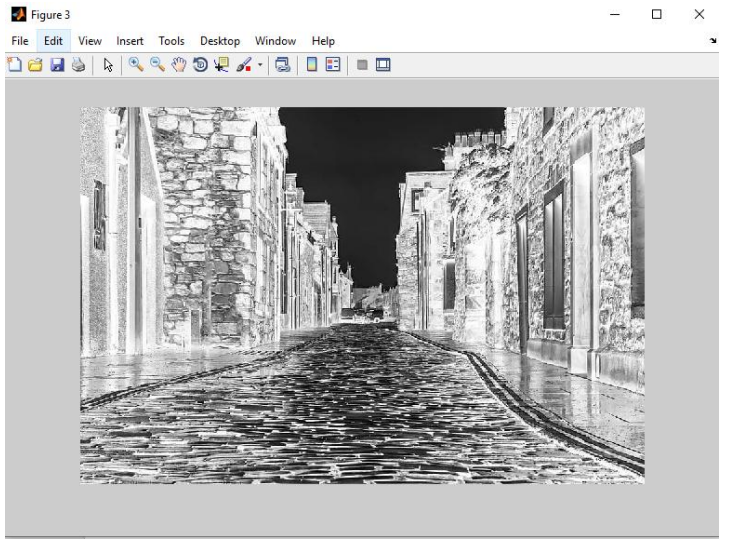


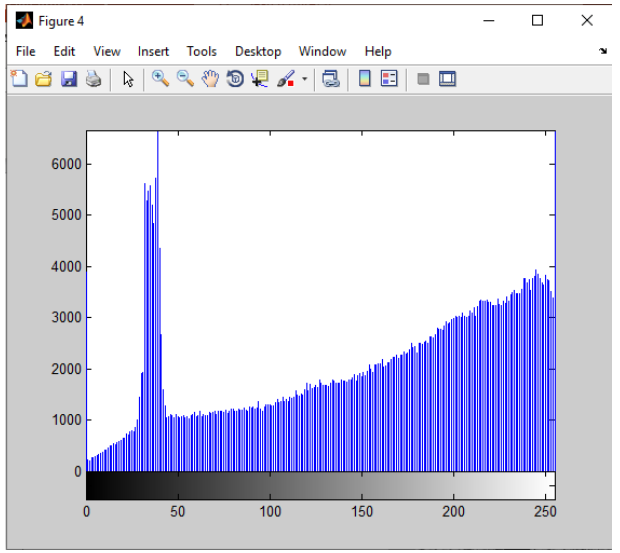



```
% Image Negative
Img_N = 256 - 1 - Img;

figure(3);
imshow(Img_N);

figure(4);
imhist(Img_N);
```

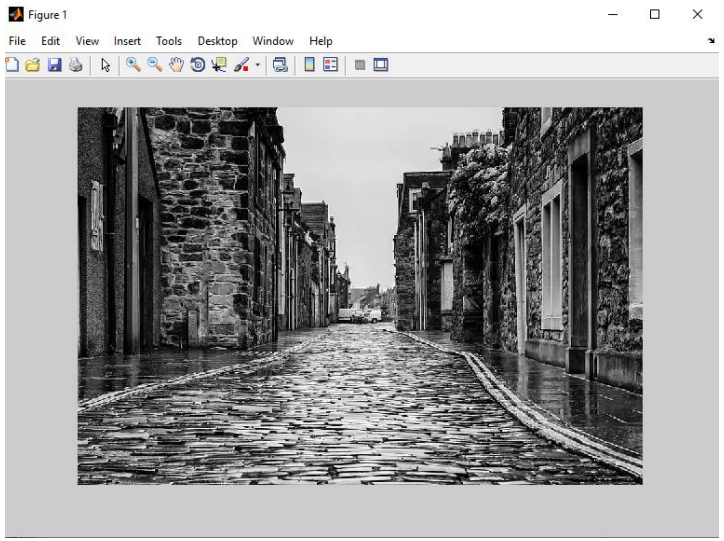


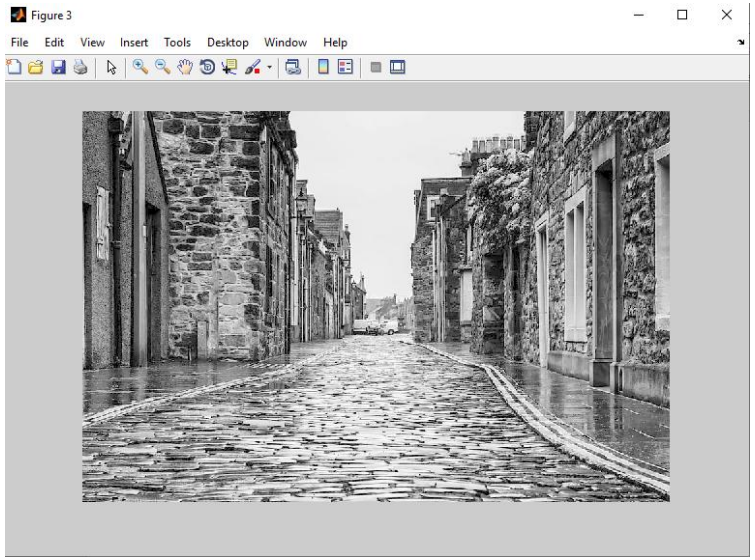


```
close(3); close(4);

% Gamma Transformation

Img_D = double(Img);
% Scale intensity values to the range [0,1]:
Img_scaled = Img_D/255;
gamma = 0.5;
c = 1;
Img_g  = c*Img_scaled.^gamma;
Img_g  = 255*Img_g;
% Scale intensity values back to the range [0,255]:
Img_g = uint8(Img_g);
figure(3);
imshow(Img_g);
figure(4);
imhist(Img_g);
```



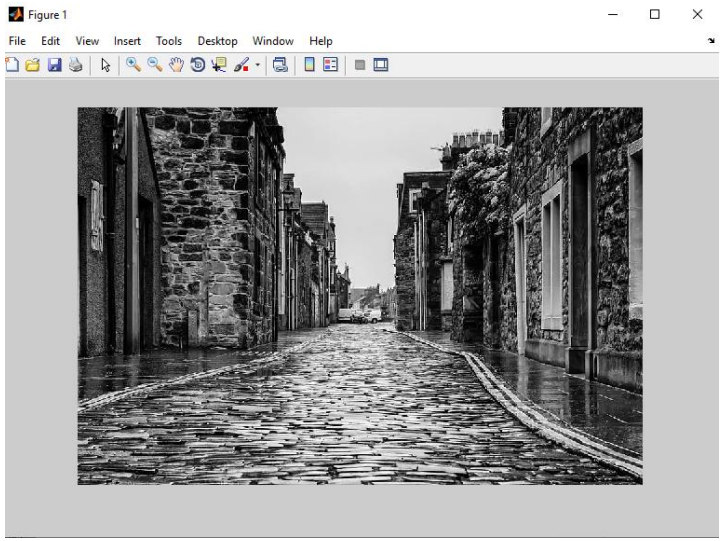


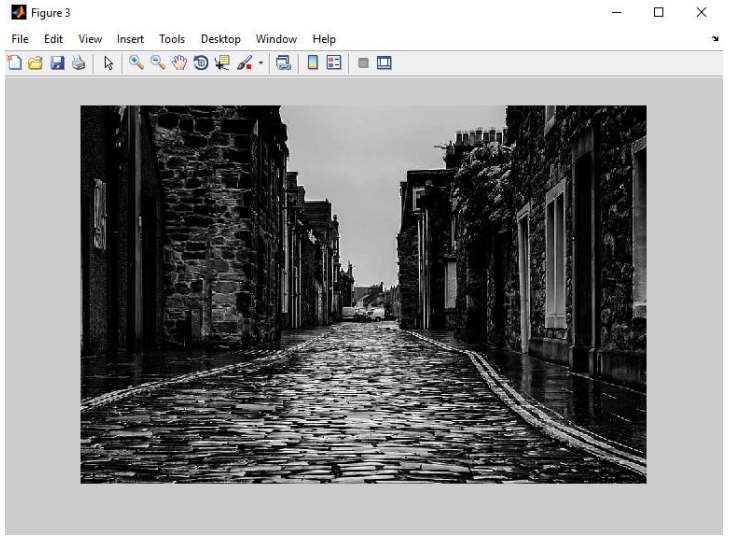
gamma = 0.5;

```
close(3); close(4);

% Gamma Transformation

Img_D = double(Img);
% Scale intensity values to the range [0,1]:
Img_scaled = Img_D/255;
gamma = 2;
c = 1;
Img_g  = c*Img_scaled.^gamma;
Img_g  = 255*Img_g;
% Scale intensity values back to the range [0,255]:
Img_g = uint8(Img_g);
figure(3);
imshow(Img_g);
figure(4);
imhist(Img_g);
```





gamma = 2;